About us

Sean Stankowski
UKRI Future Leaders Fellow

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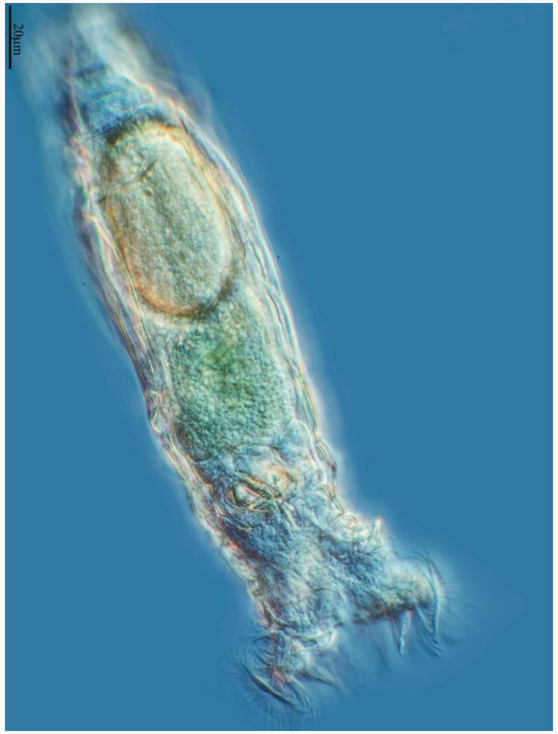
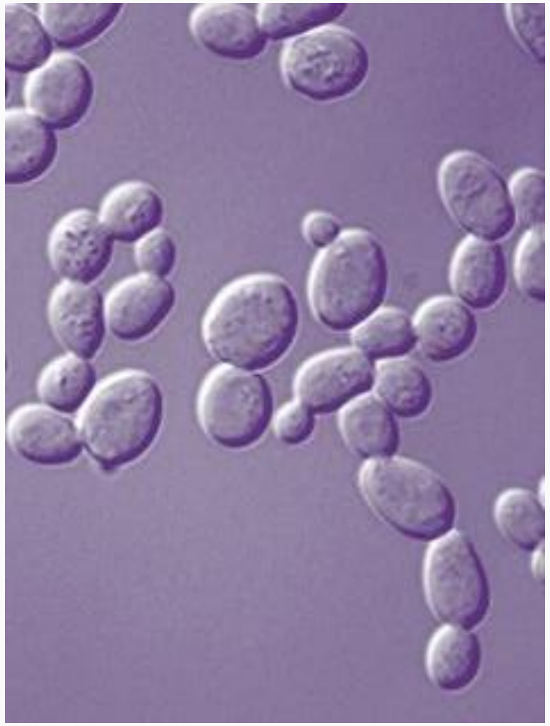
Postdoctoral Fellow

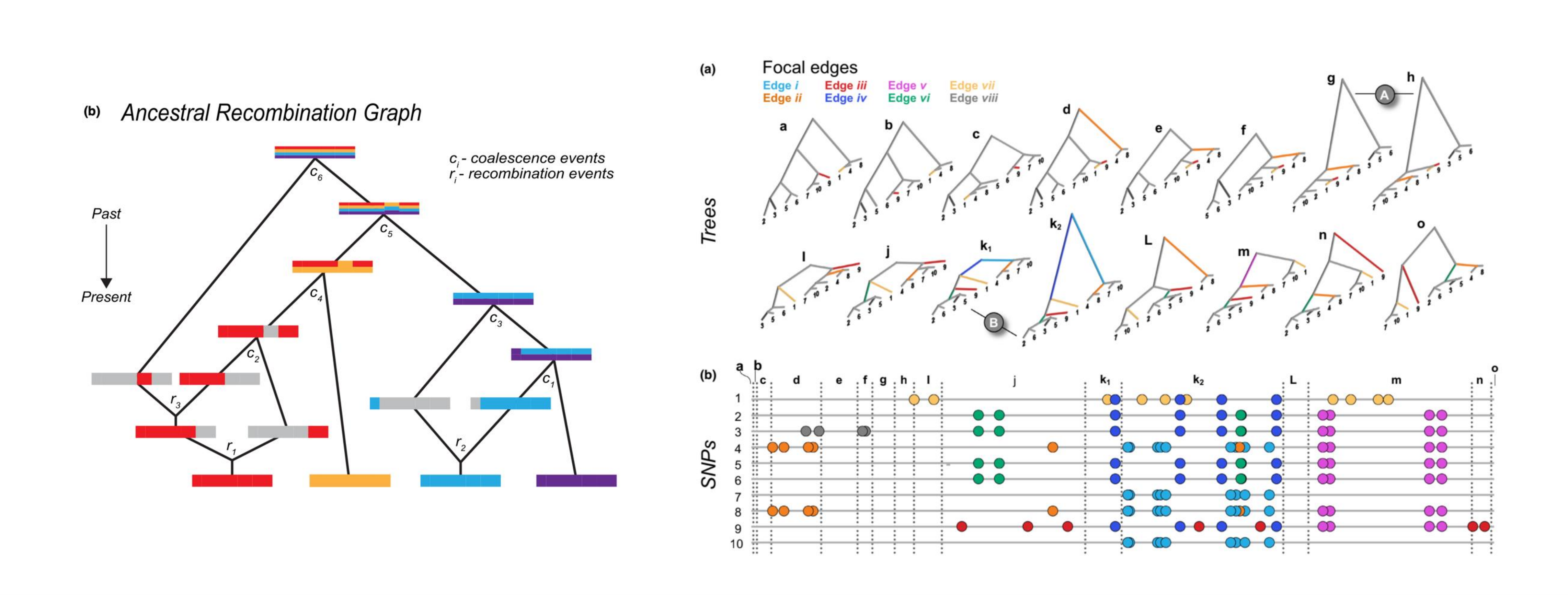
UCL Department of Genetics, Evolution, and Environment

We work with genomics data from diverse organisms





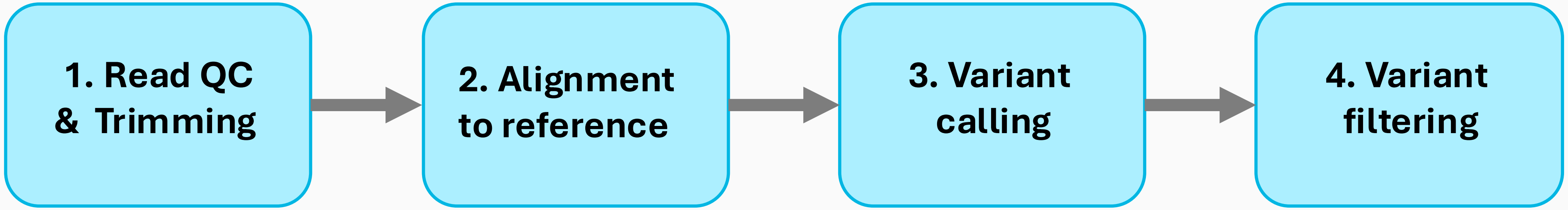
Very interested in new methods



Motivated by questions about

- Evolutionary pathways to novelty
- Adaption and speciation
- Evolution of sex and asexuality
- Genome evolution

The plan for the next 2 days



Reads to SNPs

The plan for the next 2 days

1. Read QC & Trimming → 2. Alignment to reference → 3. Variant calling → 4. Variant filtering

Reads to SNPs

Principal Component Analysis → Genome scans

Basic analysis

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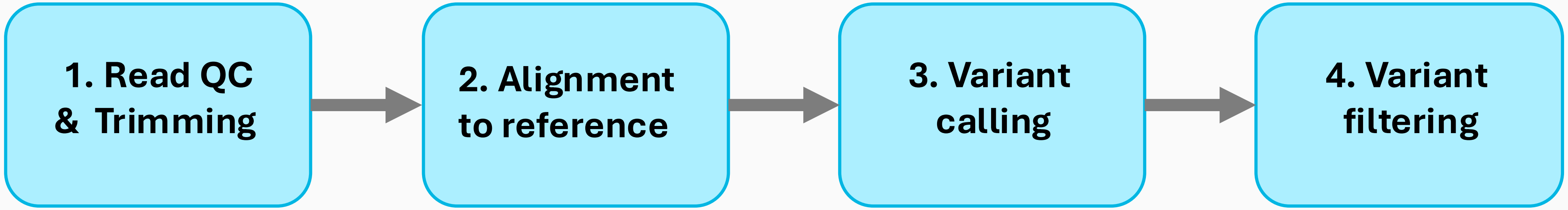

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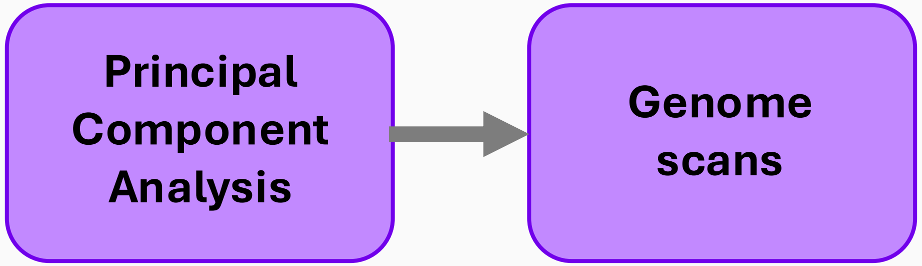
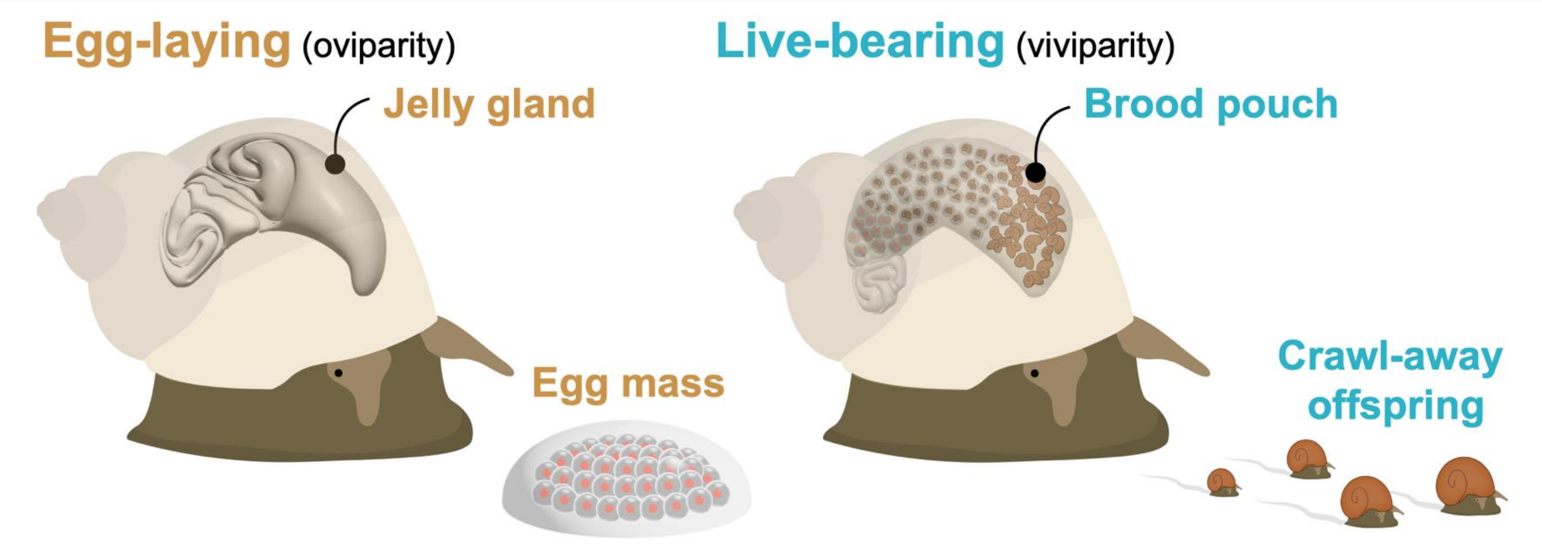


Basic analysis

But we are at your service!

The dataset we will work with

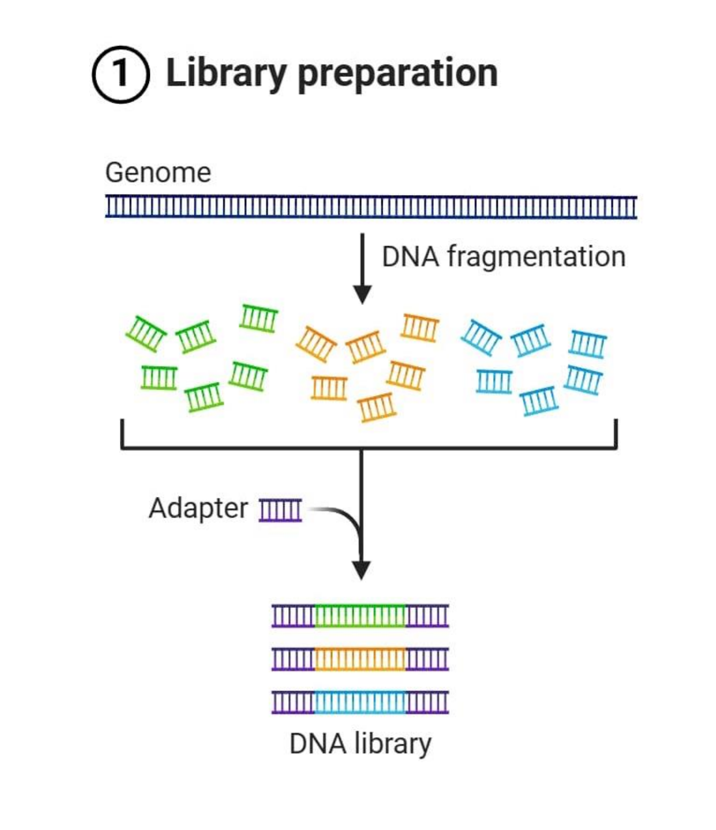
- Intertidal snails from the genus *Littorina*
- Four taxa that differ in their reproductive mode
- 108 individuals Illumina sequenced to 15x coverage



https://www.science.org/doi/10.1126/science.adi2982

Background:
Short-read sequencing and sequencing strategies

Illumina sequencing



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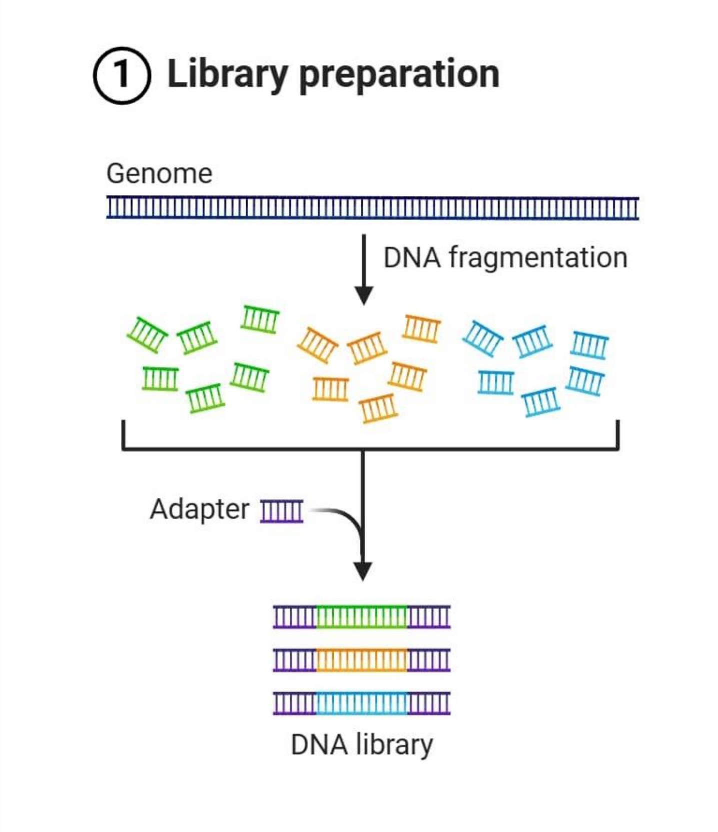
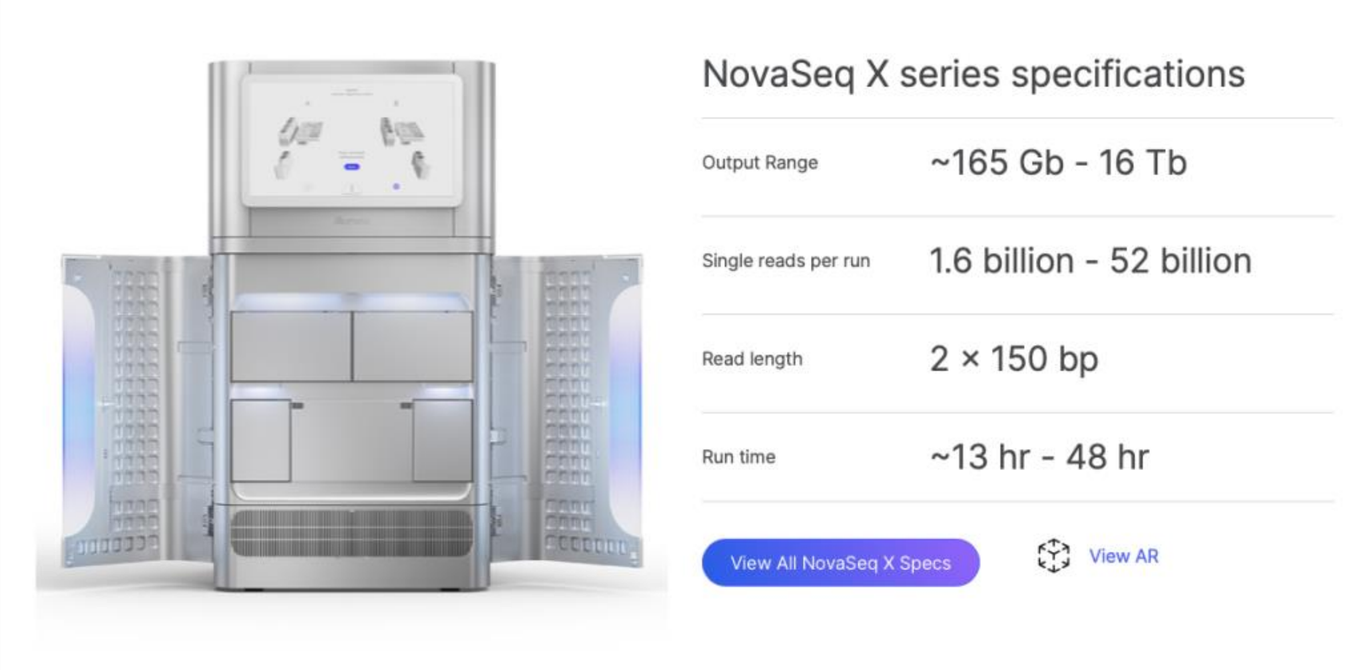
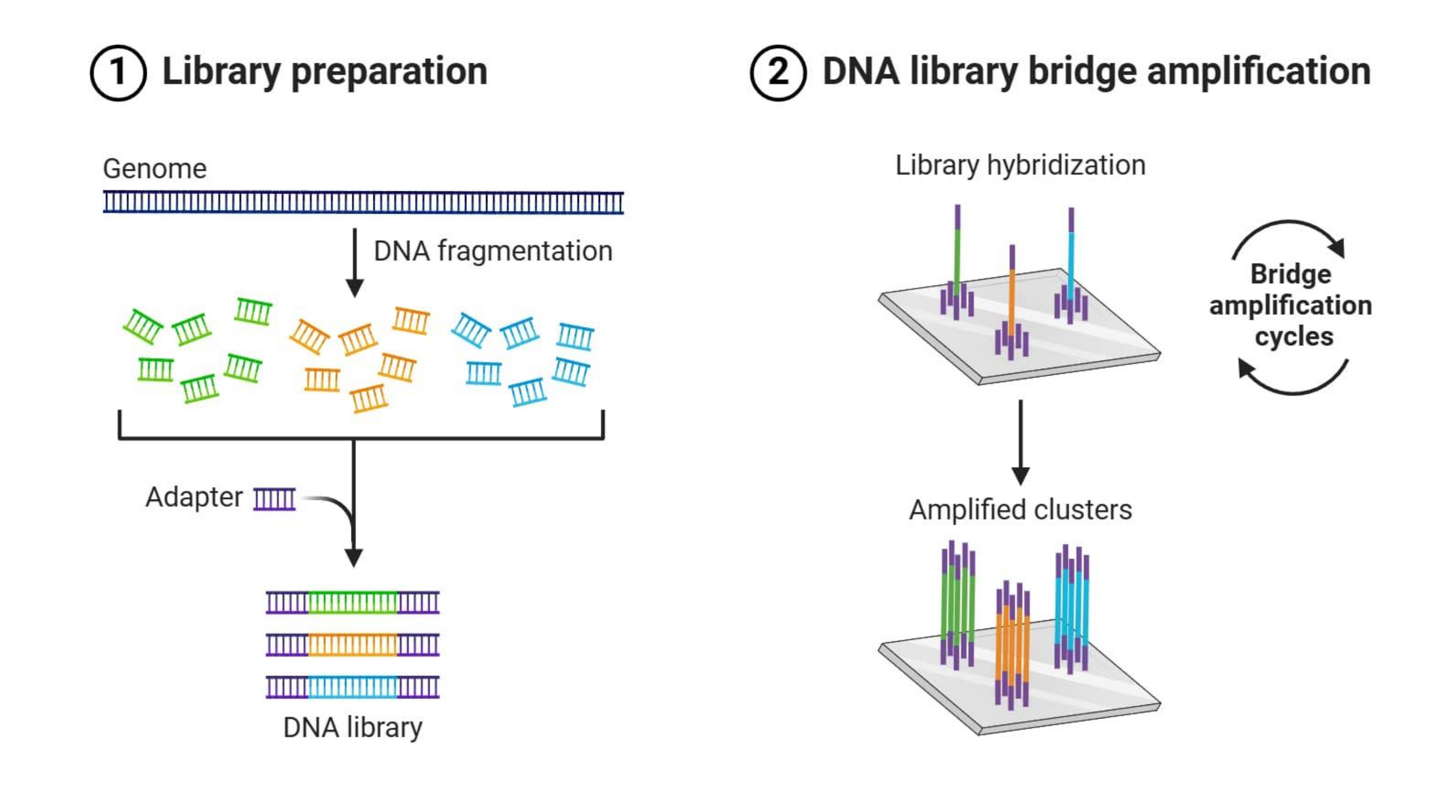




450 human genomes at 10x coverage per run

Illumina sequencing



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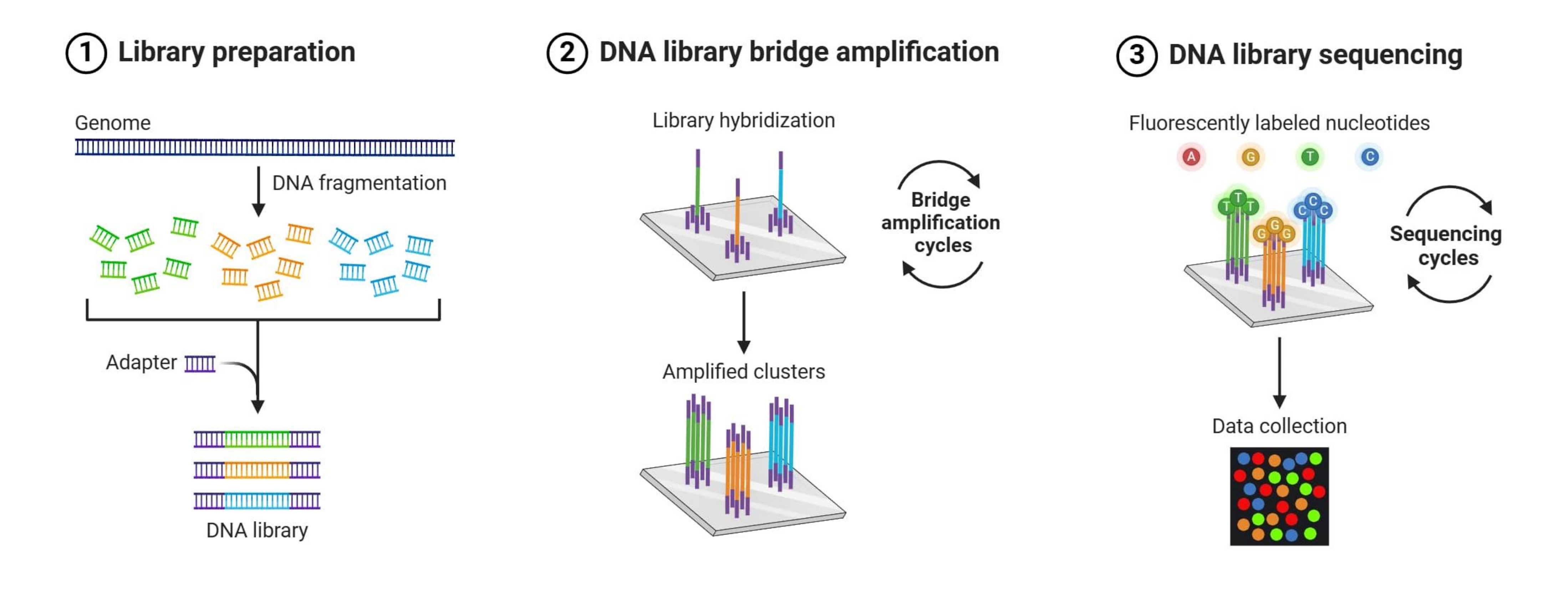

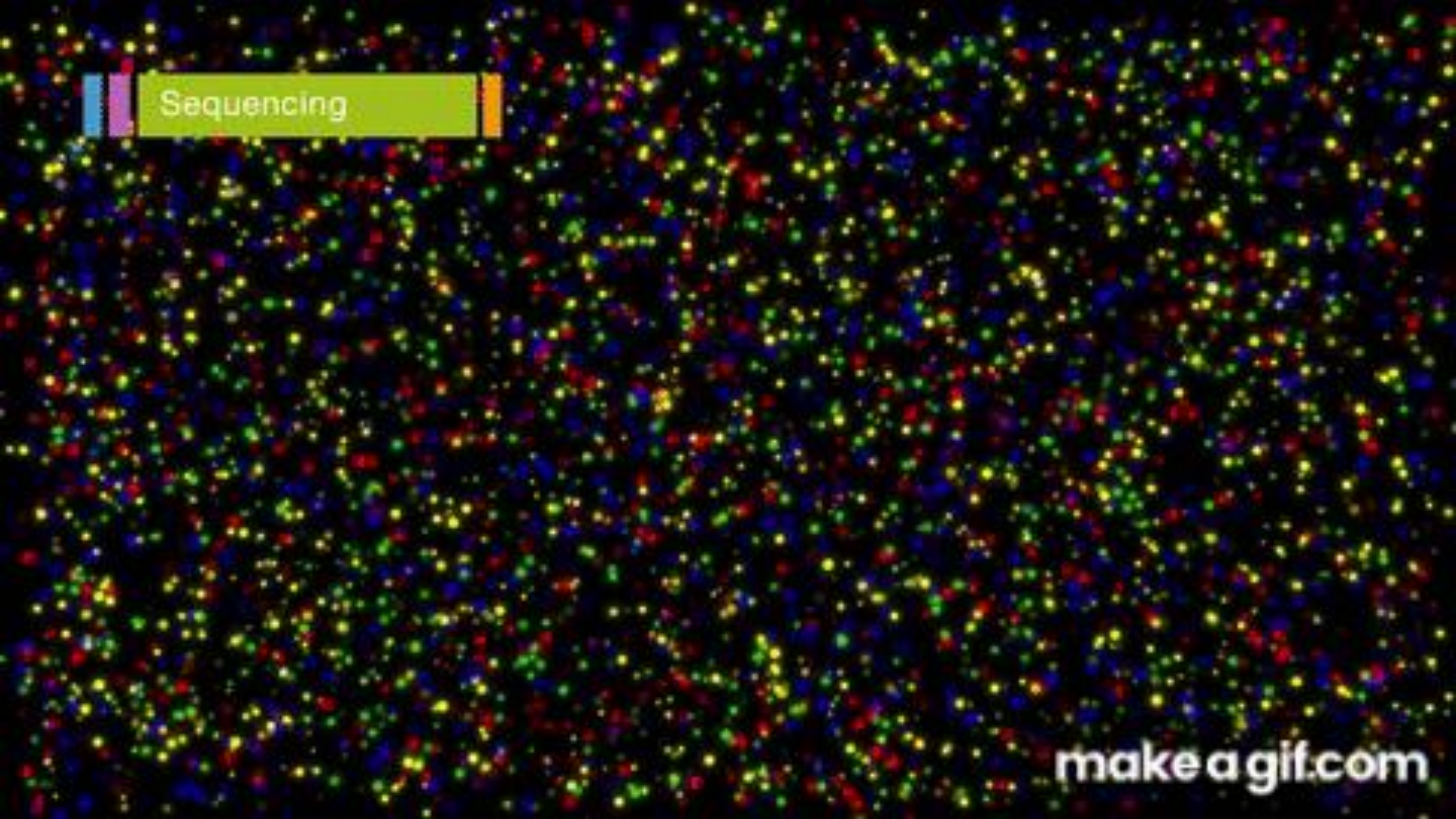
Sequencing
makeagif.com

Different sequencing strategies

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- **Linked-read sequencing:** Uses barcoded short reads derived from the same long DNA molecules to recover long-range information, including haplotypes and structural variants.

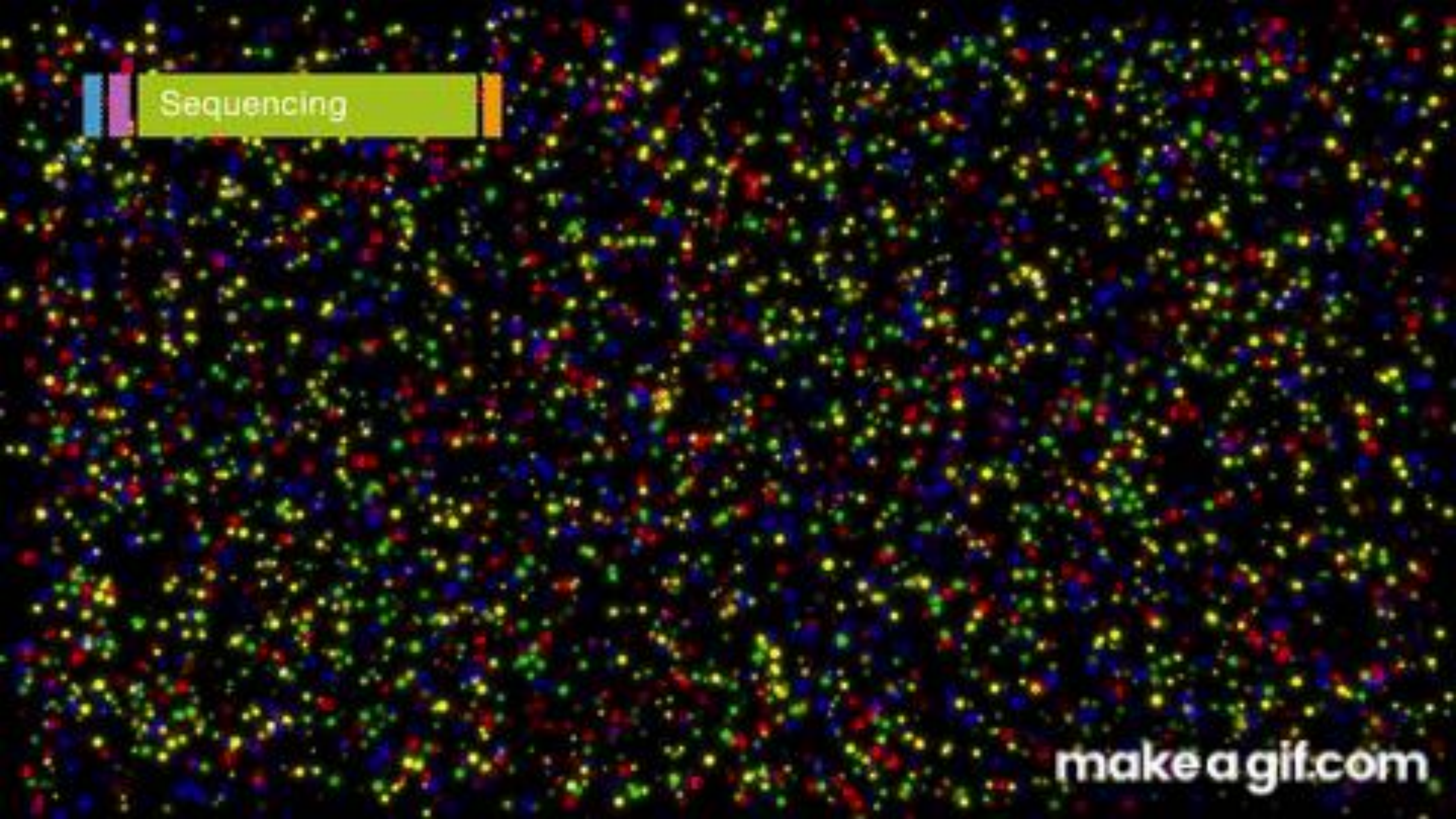
Different sequencing strategies

Approach	Cost/sample	Sample size	Genome coverage	Genotypes	Haplotypes	Main trade-off
High-coverage WGS	High	Low–medium	Whole genome; deep	High confidence	Mostly statistical phasing	Very reliable calls, high cost
Low-coverage WGS	Low–medium	High	Whole genome; shallow	Likelihoods or imputation	Statistical phasing	Broad coverage, less certainty
Pool-seq	Very low	Very high	Whole genome; pooled	No individual genotypes	None	Cheap allele frequencies, no individuals
RAD-seq	Low	High	Small subset	At sampled loci	Local only	Cheap markers, most loci missed
Linked reads	Medium–high	Low–medium	Whole genome	High confidence	Long-range phasing	Strong phasing, specialised DNA/library

Different sequencing strategies

Main objective	**Usually best suited approach**
Most accurate genome-wide genotypes for each individual	High-coverage Illumina WGS
Genome-wide population data from many individuals	Low-coverage WGS
Allele-frequency comparisons using the largest possible number of individuals	Pool-seq
Population structure or phylogeography on a restricted budget	RAD-seq
Long-range haplotypes and improved structural-variant detection using short reads	Linked-read sequencing

1.1 Quality control and trimming of raw short-read sequence data


Sequencing
makeagif.com

FASTQ Format

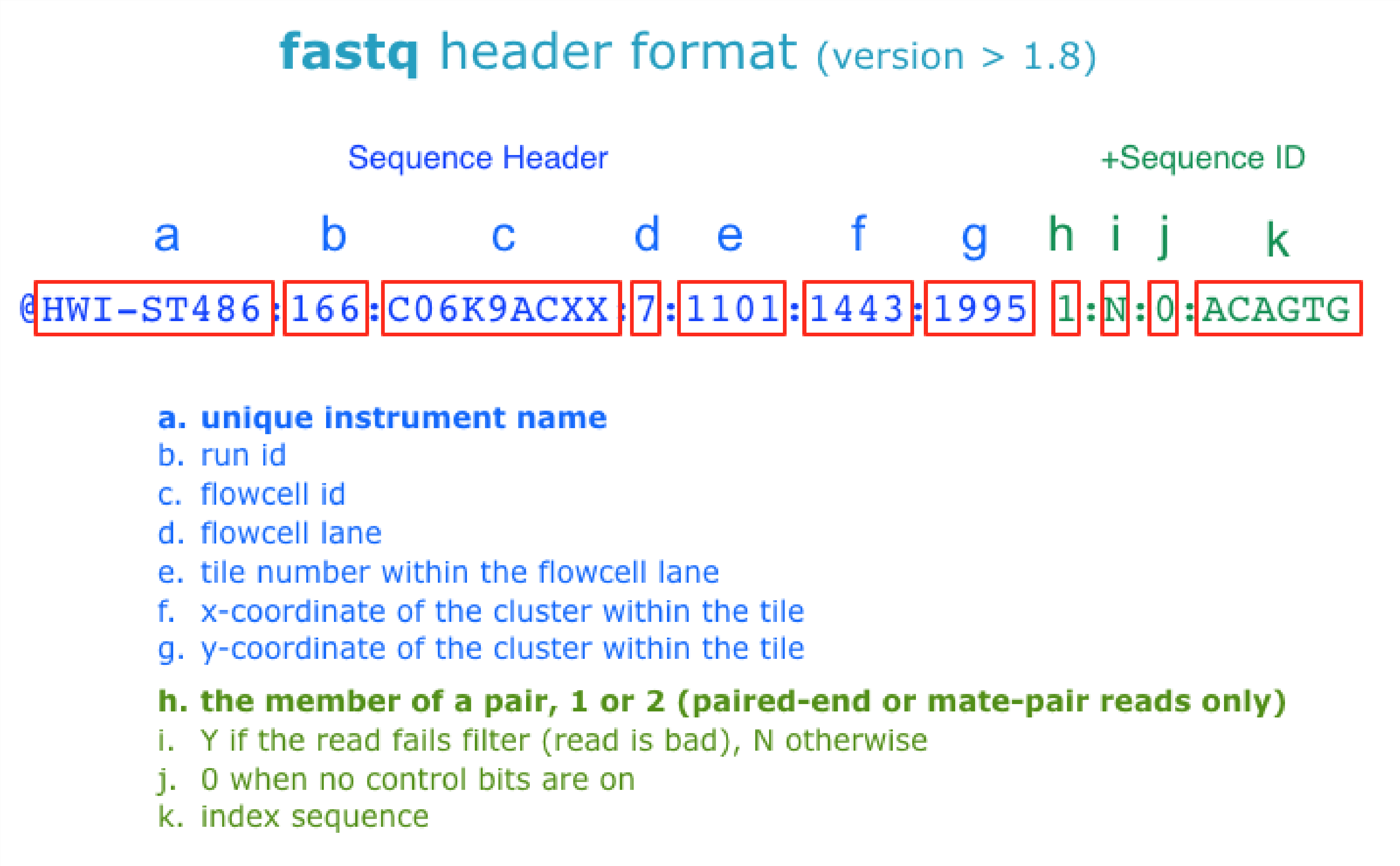
```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGAACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAGTATTTACGCCTATCCTTCATATTTTCCGCAAGGTAACTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-F-FFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFF
```

FASTQ Format

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCG
TGCAGCAGCTAATGAGGAACCACTTCCTCCCTCCAGCCGCTCTAAATACCTCAGAACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAAGAAGAAAGCACAAGTATTTACGCCTATCCTTCATATTTTCCGCAAGGTAACTATCTCGGTTTCATATCGAGATTTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGGAAGTTATGCAGGGGCATCCTGTATTATTAAATAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-F-FFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCCAGTCAATATAAGG
+
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFFFFFFF-FFFFFFFFFF
```

`@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979`	Header
`GGAGGAAGGTCCTCGCTCCTCTTTCATATAAGGAAATGGCTGAAT`	Sequence
`+`	
`FFFFHHHHHHJIJJJJJJJJIJJJIGIGIGGIJJIJIJJJJJJIII`	Quality string

FASTQ Header



FASTQ Sequence

- Contains the nucleotide sequence generated by the sequencer.
- Bases are normally represented by A, C, G and T.
- N indicates that the sequencer could not confidently identify the base.
- The sequence is the read as produced by the instrument and **may still contain adapters or other technical sequence**.

Adapter

```
GTGCCAGCCGCCGCGGTAATACGAAGGGACCTAGCGTAGTTCGGAATTACTGGGCTTAAAGAGTTCGTAGGTGGTTAAAAAAGTTGGTGGTGAAATCCCAGAGCTTAACTCTGGAACTGC
CATCAAAACTTTTTAGCTAGAGTATGATAGAGGAAAGCAGAATTTCTAGTGTAGAGGTGAAATTCGTAGATATTAGAAAGAATACCAATTGCGAAGGCAGCTTTCTGGATCATTACTGAC
ACTGAGGAAC
+
DDDDDIIHIIIIIIIIIIIIIIIHIIIIIIIIIIIIIIIIHHIIHHHHIIIIIIIIIIIIIHIIIIIHFHIIHIIIIIHGHHHIIIIIIIIIIIEHHIIHIGHIIIFEHIIIIIEGHIII
IHFHIIHHHIIIIIIHII?GHHHGHIFHHHHIIHIIIIIIHIIIIIIIIIIHIIGHGIIIIIIIIHIHIIIFHIIIIIHIIIIGHGHIHHHIIIIHHHIIIIHHIIIHHHGHIHIHHHHH
HHIIIIIGFH
```

Phred+33 quality-scores encoding

Phred score	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Character	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5
Phred score	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41
Character	6	7	8	9	:	;	<	=	>	?	@	A	B	C	D	E	F	G	H	I	J

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Character	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5
Phred score	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41
Character	6	7	8	9	:	;	<	=	>	?	@	A	B	C	D	E	F	G	H	I	J

Phred score	Error probability
Q10	1 in 10
Q20	1 in 100
Q30	1 in 1,000
Q40	1 in 10,000

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Character	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4	5

Phred score	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41
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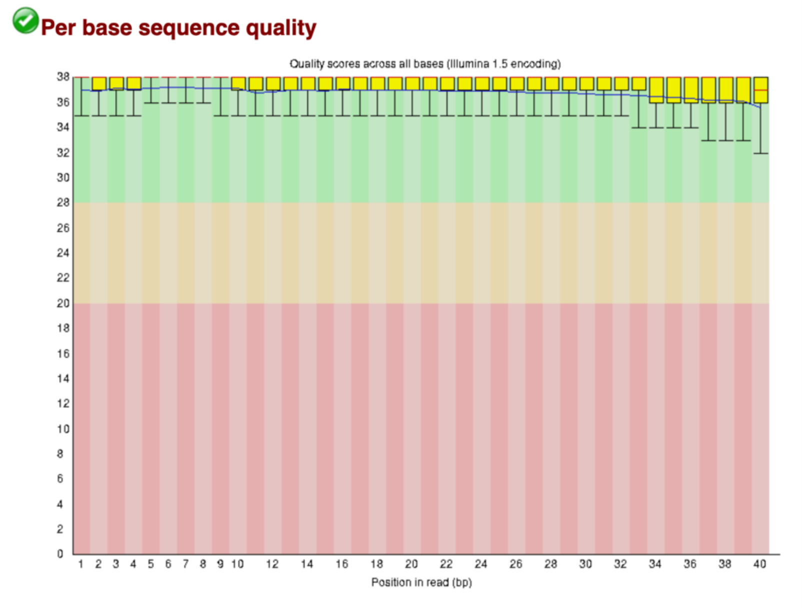
```

What do we need to do with these raw data?

- Preform initial QC
  - Check the length and quality of reads etc…



Nice per base quality!

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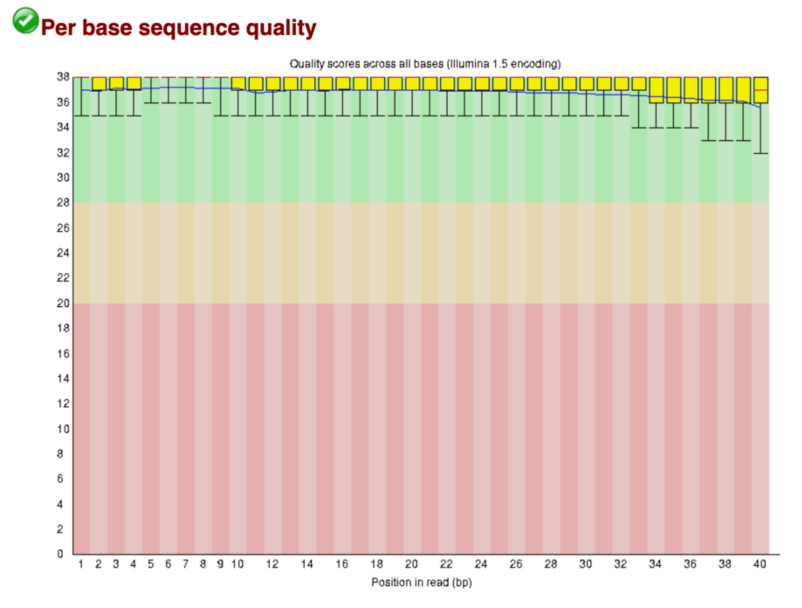

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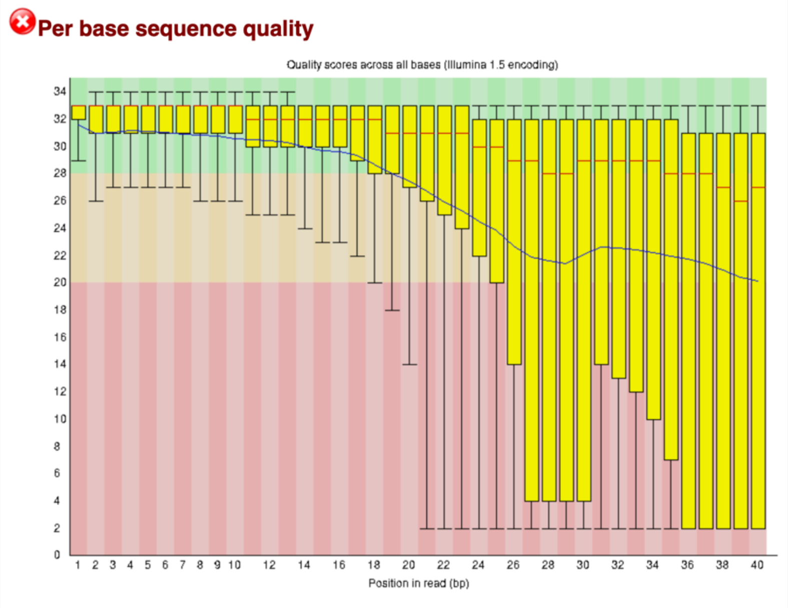

Not so nice…

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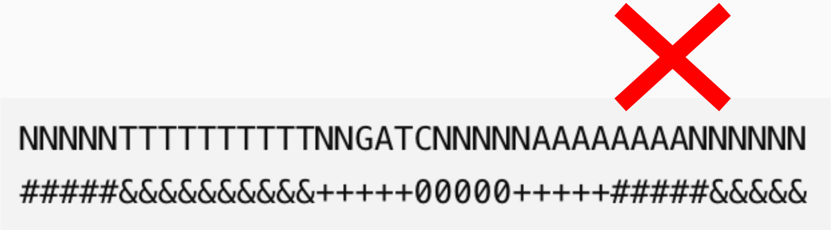
- Trim non-biological sequences and discard low-quality reads



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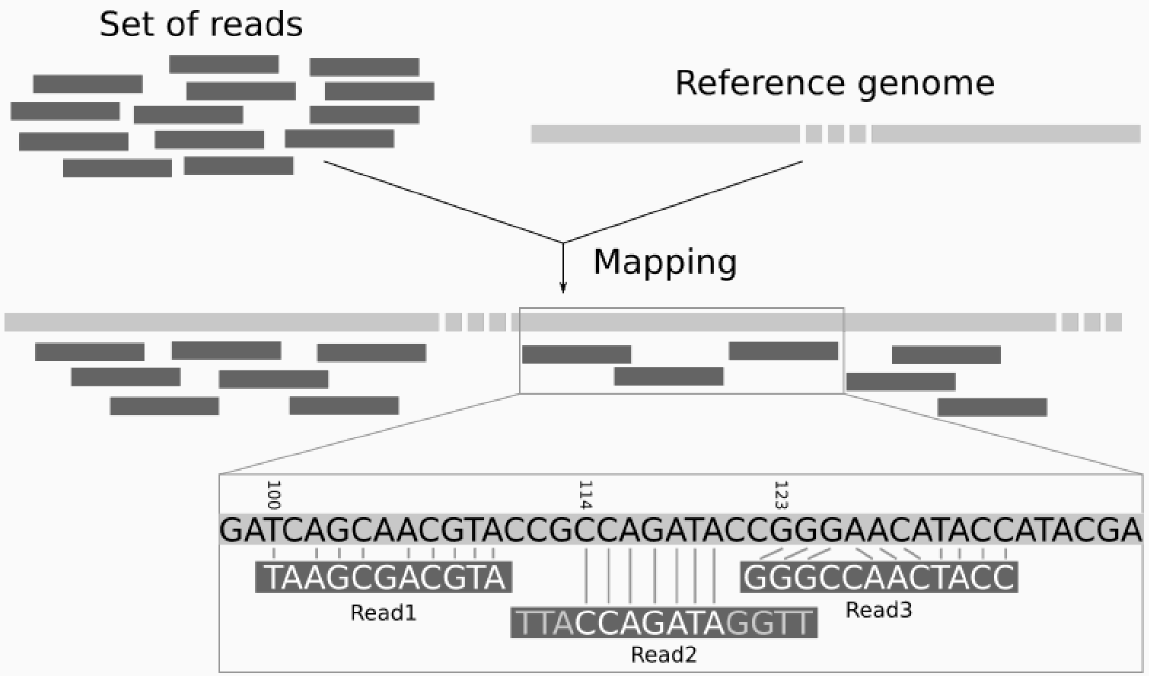
Let's practice

1.2 Mapping short-read data to a reference genome

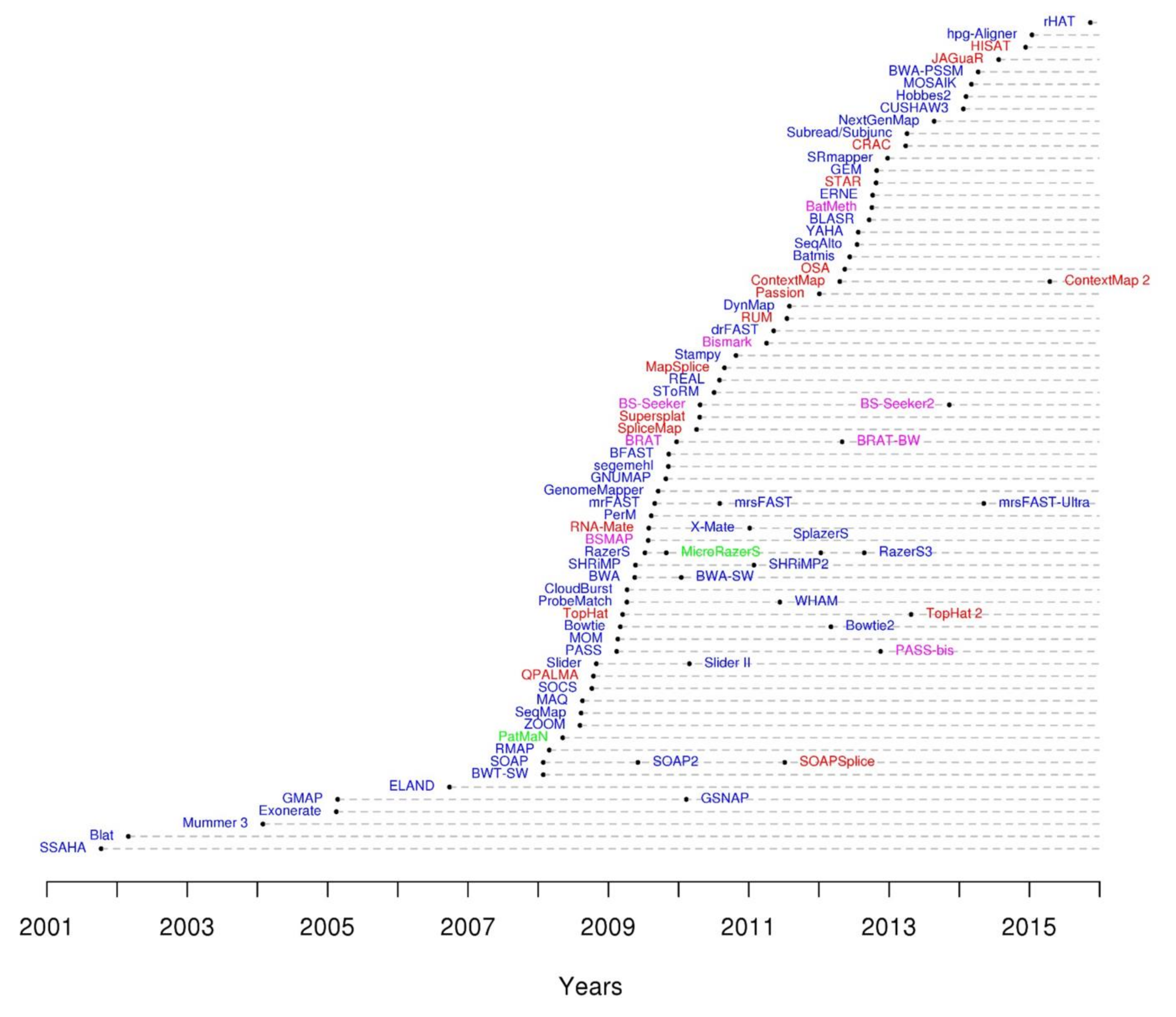
What is read mapping?

The process of finding where a sequencing read most likely originated within a reference genome

https://training.galaxyproject.org/training-material/topics/sequence-analysis/tutorials/mapping/tutorial.html



More than 90 mappers to choose from!



https://www.ecseq.com/support/ngs/what-is-the-best-ngs-alignment-software

BWA-MEM2

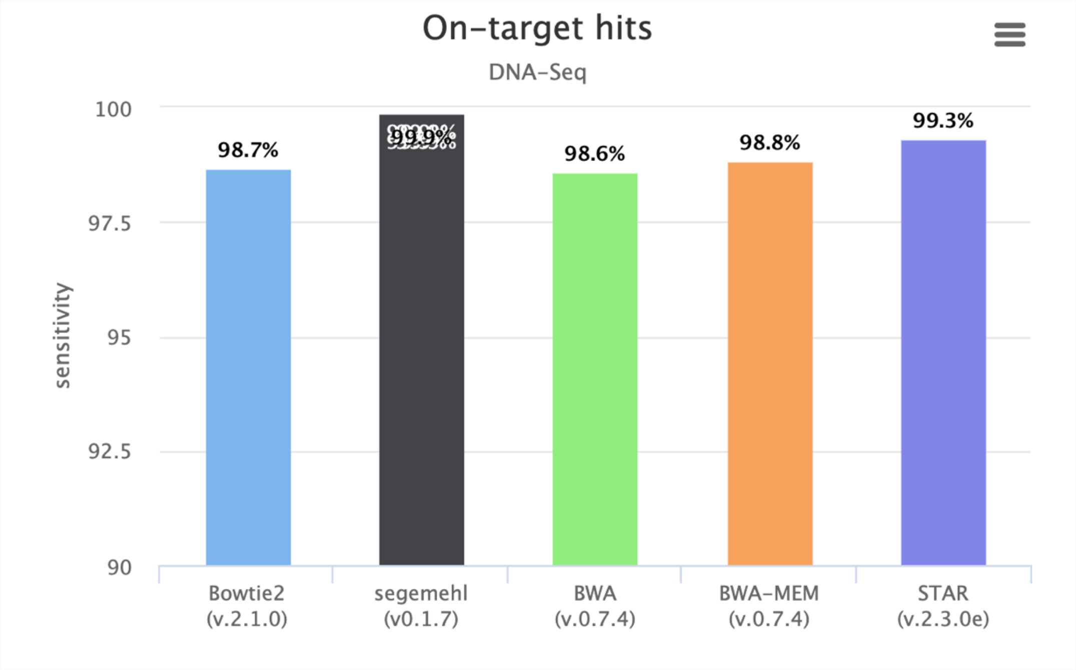
https://github.com/bwa-mem2/bwa-mem2

- Very widely used, efficient, and open source
- Leverages the Burrows-Wheeler transform
- The **Burrows–Wheeler transform (BWT)** makes short-read mapping efficient by reorganizing the reference genome into an index that can be **searched rapidly.**

https://en.wikipedia.org/wiki/Burrows%E2%80%93Wheeler_transform

https://www.youtube.com/watch?v=4n7NPk5lwbI

BWA-MEM2: performance



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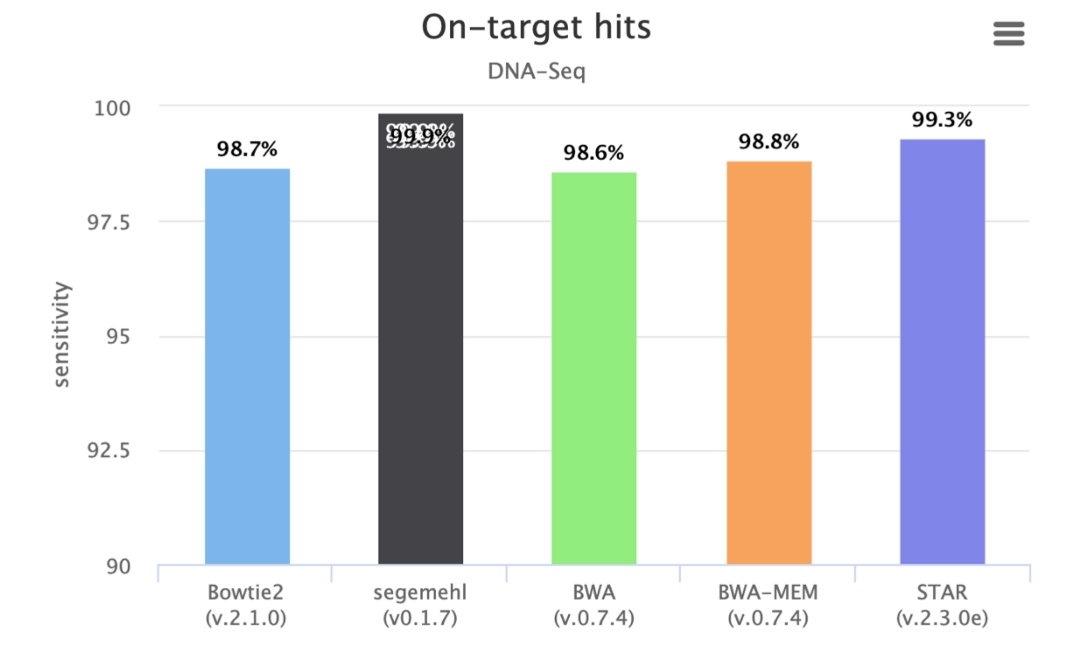

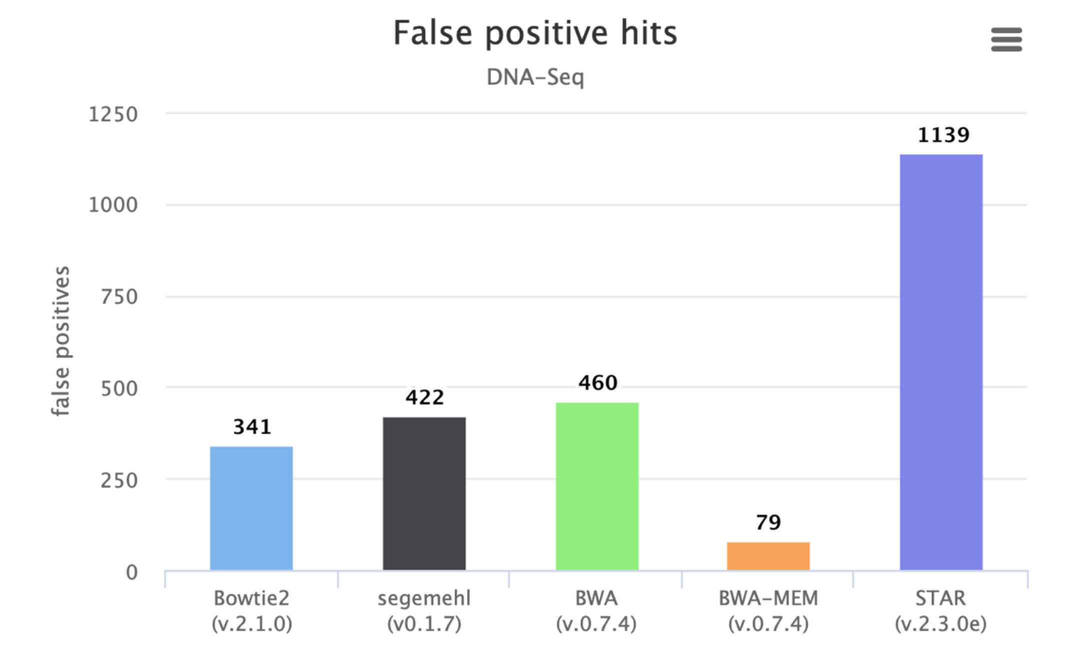

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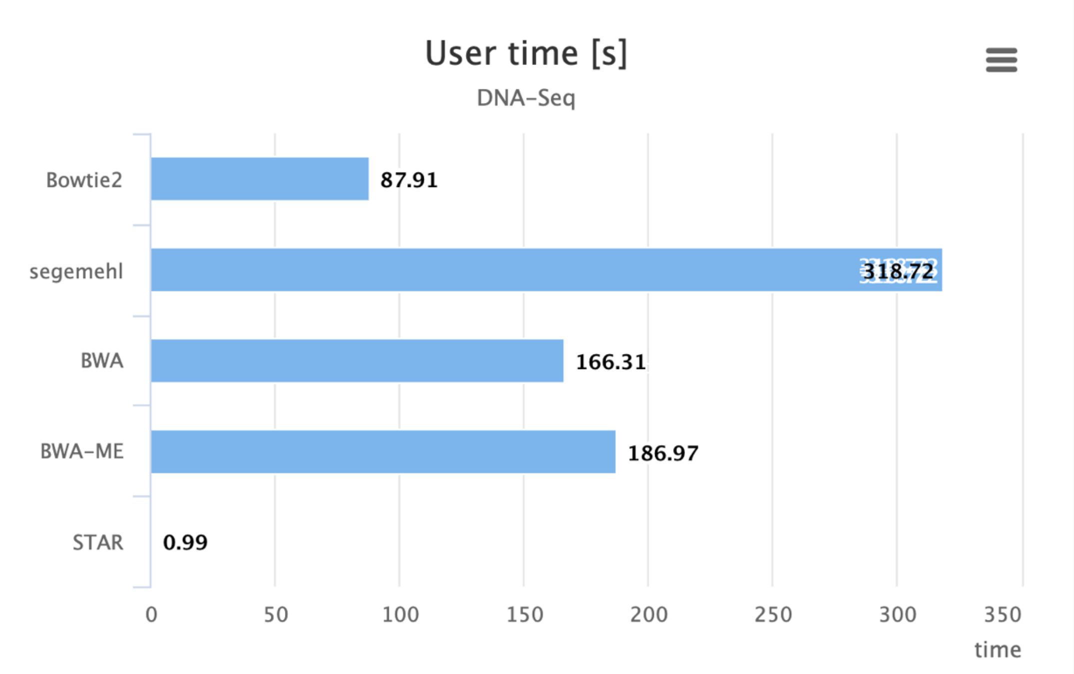

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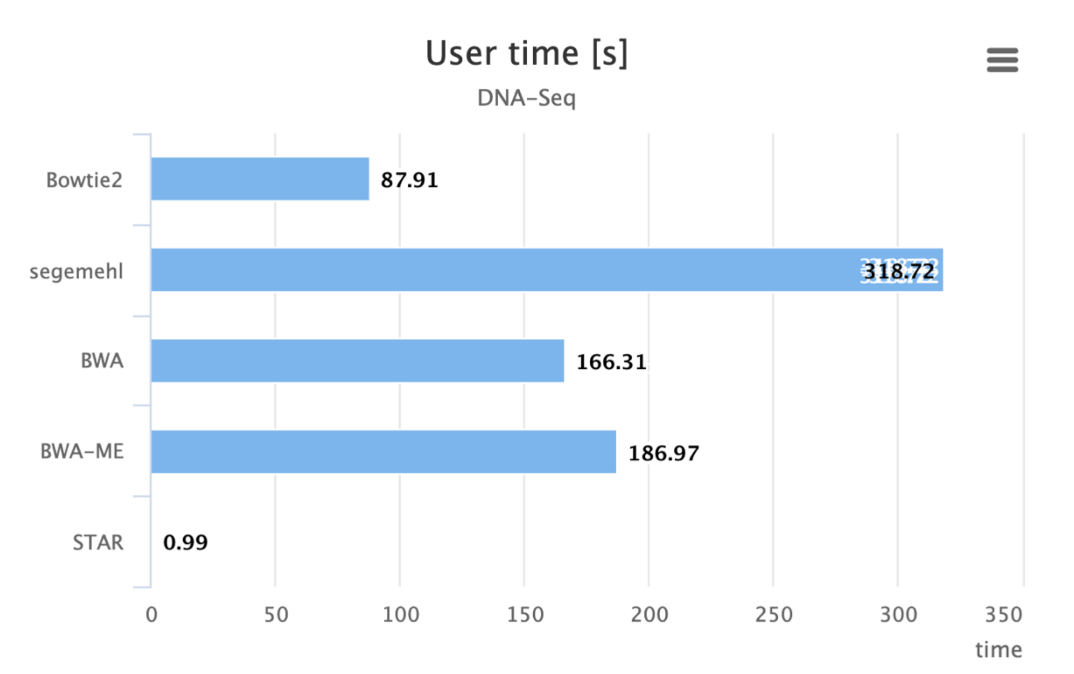

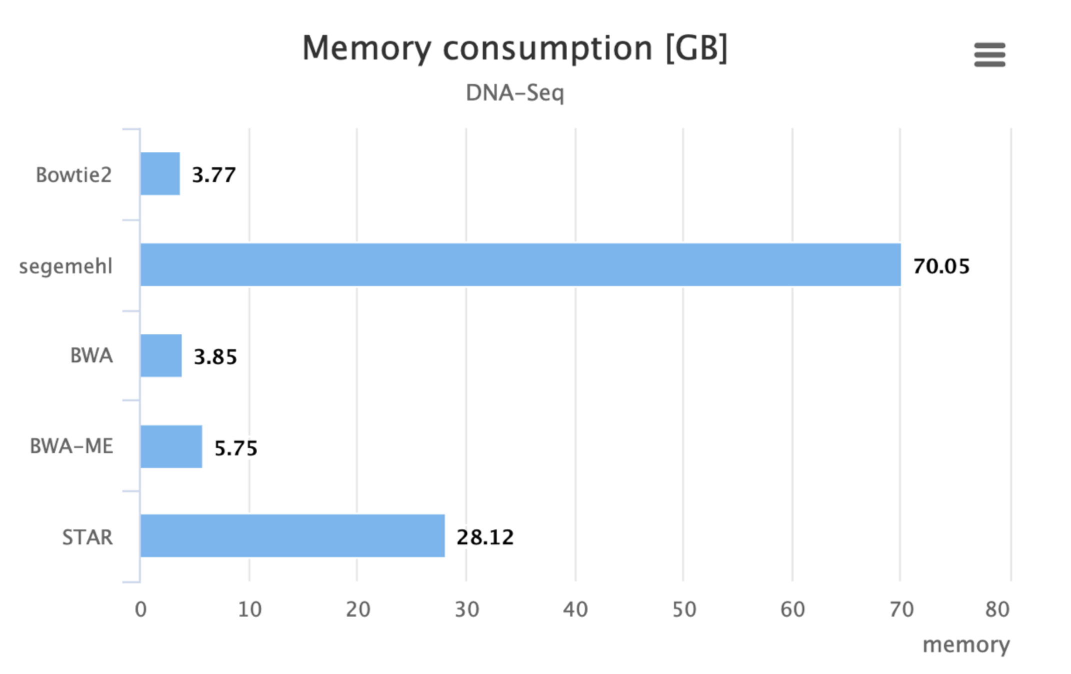

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SAM and BAM files

- **A SAM file (Sequence Alignment/Map)** is a human-readable, tab-delimited text file that records how sequencing reads align to a reference genome.
- Contains an optional header followed by one line for each read alignment, including its genomic position, mapping quality, CIGAR string, sequence and quality scores.

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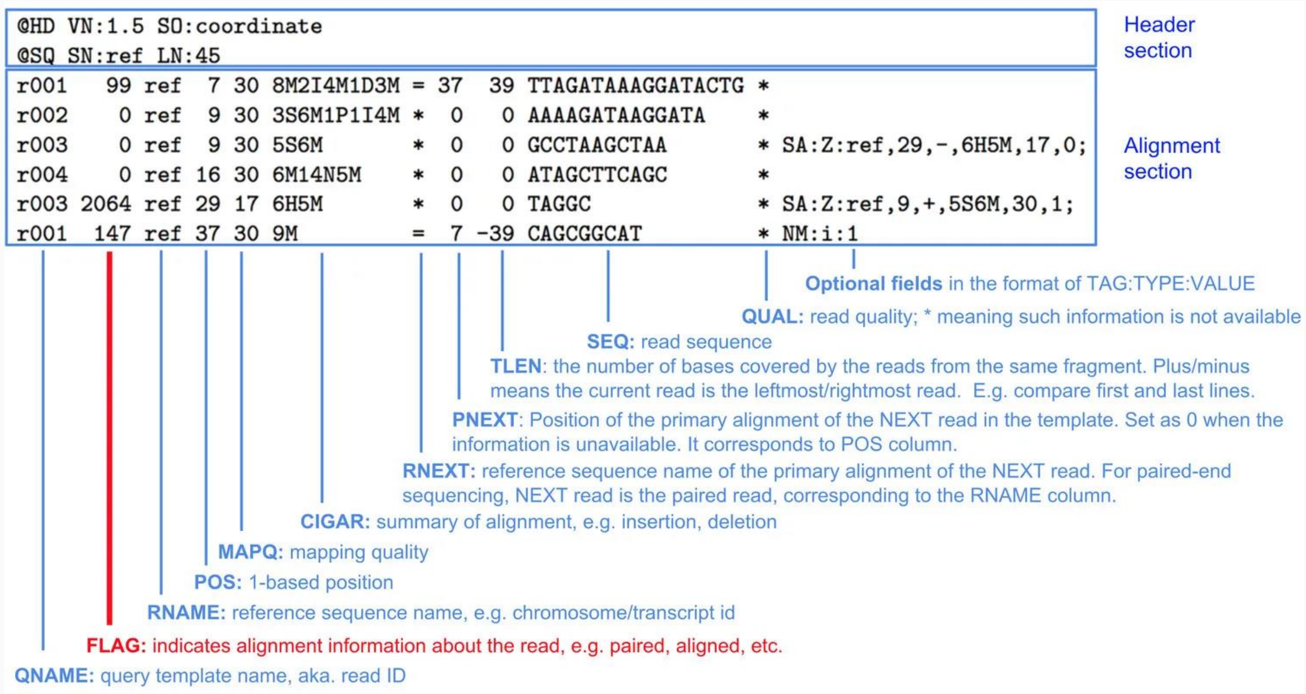

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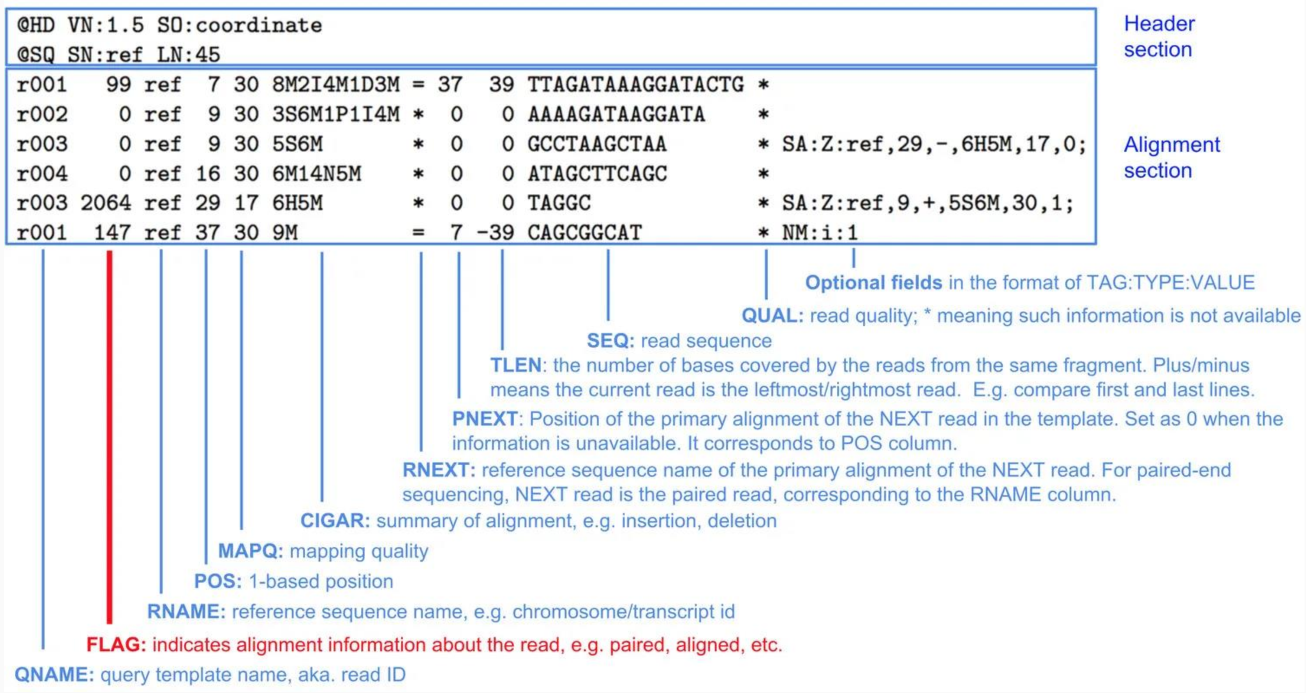
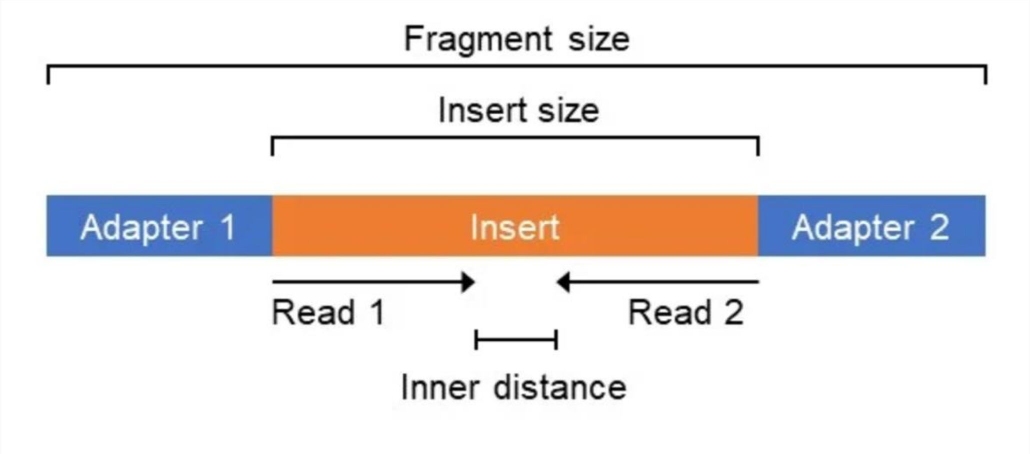


BAM (Binary Alignment/Map) contains the same information but stored in a compressed binary format.

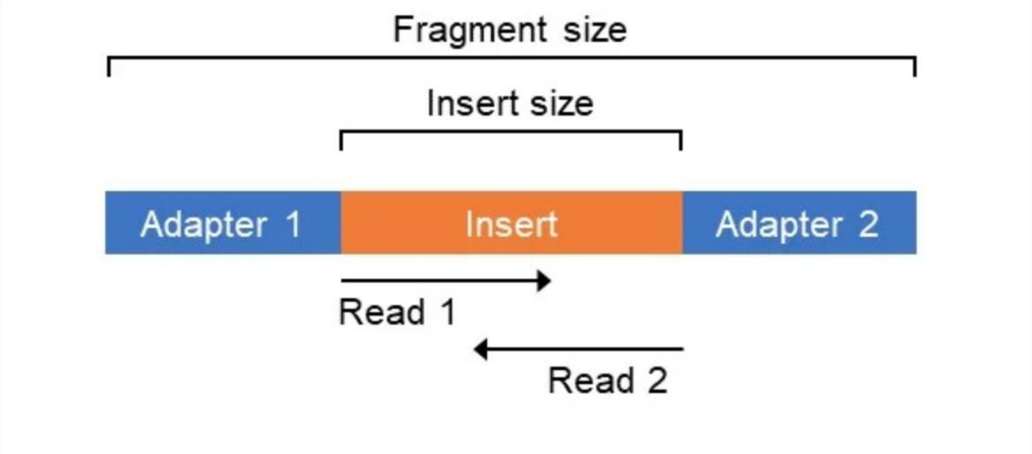
BAM files are smaller, faster for software to process and can be indexed, allowing programs to retrieve reads from a particular genomic region without reading the entire file.

Most Illumina sequences are now paired-end

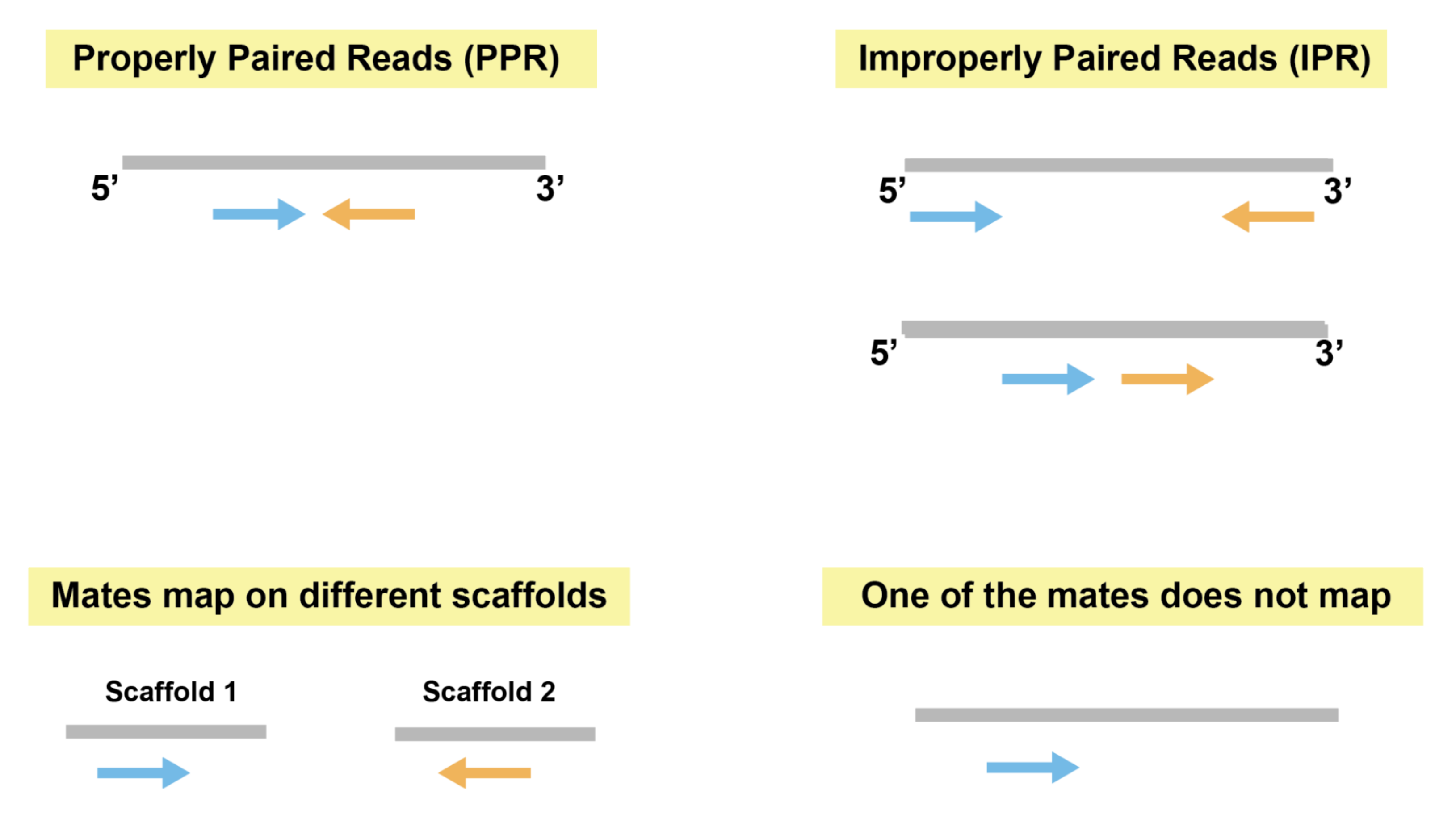
Non-overlapping PE



Overlapping PE



Paired reads allow us to assess mapping

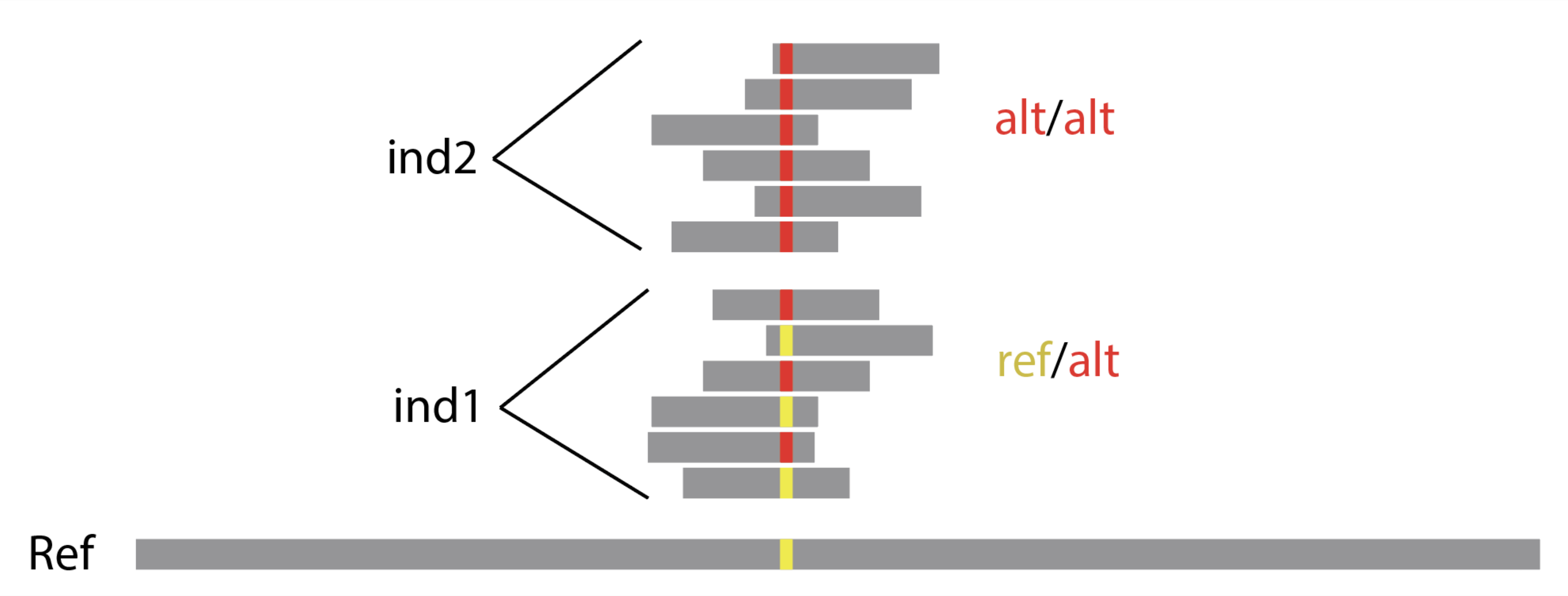


1.3 Multi-sample variant calling with BCFtools

Variant calling

The process of inferring genetic differences from sequencing reads relative to a reference genome:

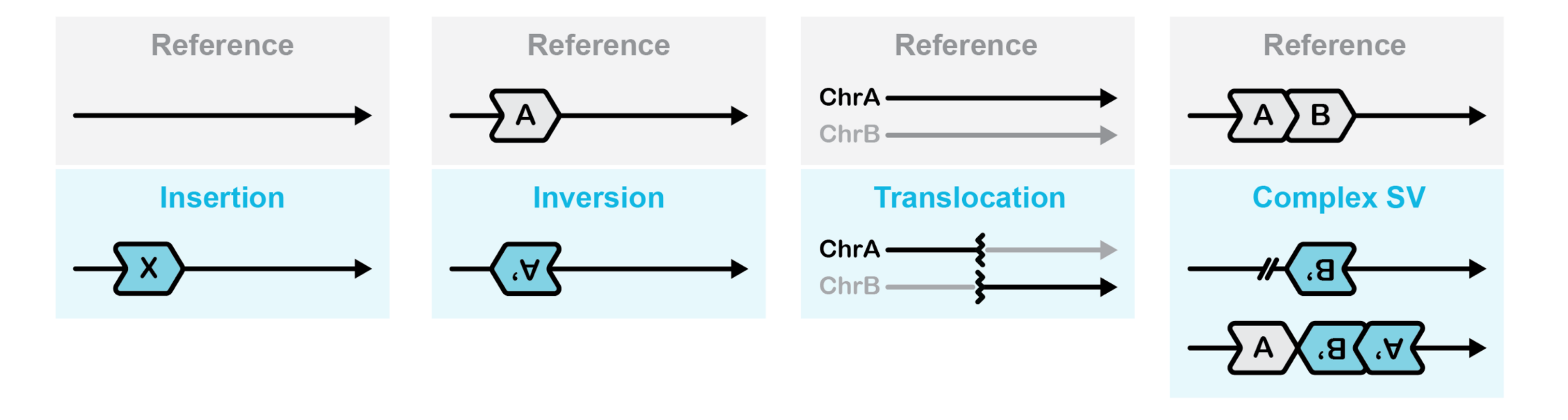
A single-nucleotide polymorphism (SNP)



Variant calling

The process of inferring genetic differences from sequencing reads relative to a reference genome:

Structural variants also!



Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from n reads.

Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from *n* reads.

How random flips would you want to confidently conclude that this is not a 2-headed coin?

The reality is that you can never be 100% certain

Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from *n* reads.

$$P_{\text{both alleles}} = 1 - 2^{1-n}$$

where n is the number of reads

```
reads <- 1:20
prob_both <- 1 - 2^(1 - reads)
data.frame(
reads = reads,
probability = prob_both,
percentage = 100 * prob_both
)
```

Reads	Probability of observing both alleles
1	0.00%
2	50.00%
3	75.00%
4	87.50%
5	93.75%
6	96.88%
7	98.44%
8	99.22%
9	99.61%
10	99.80%
11	99.90%
12	99.95%
13	99.98%
14	99.99%
15	99.994%

Variant calling

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(1-93.75) * 6,500,000 = 406,250...

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- The caller combines the evidence from the sequencing reads with the expected frequency of each genotype to estimate which genotype is most likely.

- The effect of the prior is strongly dependent on the number of reads that support a genotype.

Its not all about variants!

- The default for most callers is to only write variant sites to the output
- Many analyses require information from both variant and invariant sites

#CHROM	POS	Ind1	Ind2	Ind3	Ind4	Ind5	Ind6
Contig1	100	0/0	0/0	0/1	0/1	1/1	1/1
Contig1	110	1/1	0/1	0/0	0/0	0/1	1/1

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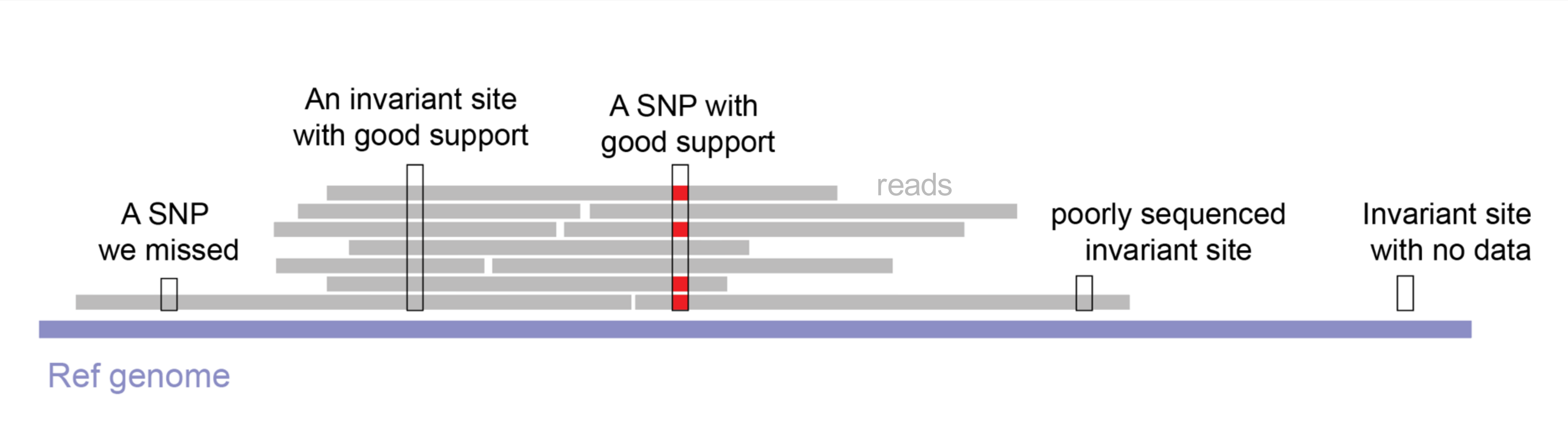

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- Many analyses require information from both variant and invariant sites
- We cannot assume that sites not called as variants are invariant
- e.g., a site not in our variant-only VCF may have been poorly sequenced, rather than showing strong evidence for being monomorphic
- The same level of evidence must be used for identifying variant or invariant sites
- For many caller there is an option to emit all sites (i.e., not just variants)—use this or live to regret it!

Many callers are available!

They have very different philosophies!

- **Pileup callers:** Evaluate each site independently using the reads aligned to it; fast but less effective for complex variation.

- **Haplotype callers:** Reconstruct local sequences to identify SNPs, indels and complex variants.

- **Machine-learning callers:** Learn to distinguish variants from errors using validated training data.

- **Genotype-likelihood approaches:** Retain uncertainty in each genotype rather than making hard calls.

Many callers are available!

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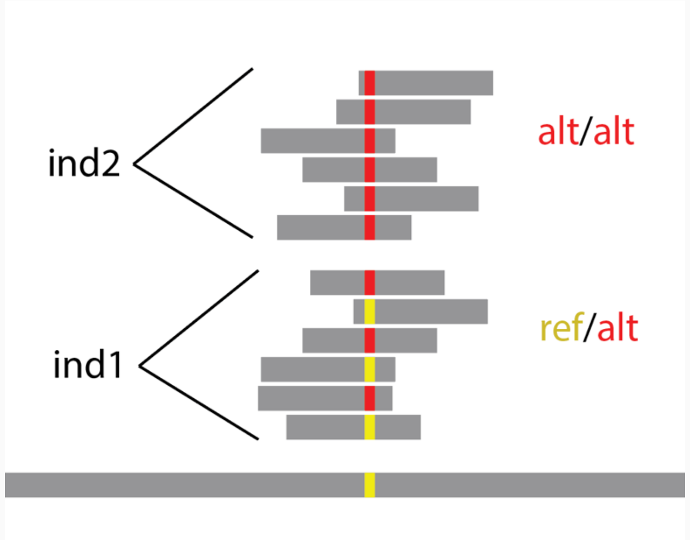
BCFtools

mpileup | call

BCFtools mpileup | call

- **Each base called independently of all others**
- **Builds a pileup:** Collects reads aligned to each position in the reference genome.
- **Weights read evidence:** Uses base and mapping qualities rather than simply counting alleles.
- **Calculates genotype likelihoods:** Estimates how well the reads support each possible genotype.
- **Calls variants across samples:** By default, treats samples as one population and applies an HWE assumption (can be deactivated using –G option)



- **Reconstructs haplotypes:** Locally reassembles reads into candidate sequences rather than examining sites independently.

- **Calculates genotype likelihoods:** Uses a Pair-HMM to determine how well each haplotype explains the reads.

- **Supports joint genotyping:** Calls samples individually as gVCFs before combining evidence across samples.

- **Much easier to combined new samples with old ones**: with others all individuals need to be recalled

gatk4

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scientific reports

Check for updates

OPEN

The evaluation of Bcftools mpileup and GATK HaplotypeCaller for variant calling in non-human species

Messaoud Lefouili & Kiwoong Nam

Identification of genetic variations is a central part of population and quantitative genomics studies based on high-throughput sequencing data. Even though popular variant callers such as Bcftools mpileup and GATK HaplotypeCaller were developed nearly 10 years ago, their performance is still largely unknown for non-human species. Here, we showed by benchmark analyses with a simulated insect population that Bcftools mpileup performs better than GATK HaplotypeCaller in terms of recovery rate and accuracy regardless of mapping software. The vast majority of false positives were observed from repeats, especially for GATK HaplotypeCaller. Variant scores calculated by GATK did not clearly distinguish true positives from false positives in the vast majority of cases, implying that hard-filtering with GATK could be challenging. These results suggest that Bcftools mpileup may be the first choice for non-human studies and that variants within repeats might have to be excluded for downstream analyses.

1.4 Filtering variants for quality

You now have genotypes! Job done?

Not quite! many of these may be unreliable due to poor sequencing coverage and others will be artifacts of thing like poor alignment

#CHROM	POS	Ind1	Ind2	Ind3	Ind4	Ind5	Ind6
Contig1	100	0/0	0/0	0/1	0/1	1/1	1/1
Contig1	200	0/0	0/1	0/1	0/0	0/1	1/1
Contig1	300	1/1	0/1	0/0	0/0	0/1	1/1
Contig1	400	0/0	0/0	0/0	0/1	0/1	0/1
Contig1	500	0/1	0/1	1/1	0/0	0/1	1/1
Contig1	600	0/0	0/1	0/0	1/1	0/1	1/1
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Contig1	800	0/0	0/0	0/1	0/1	0/1	1/1
Contig1	900	0/1	1/1	0/1	0/0	0/0	0/1
Contig1	1000	0/0	0/1	1/1	0/1	0/0	1/1

Two ways we can apply filters

Hard Filtering: The removal of entire *sites* from a vcf file

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Soft filtering of genotypes: The masking of *genotypes* from a VCF file

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If we sequence to an expected depth of x, we expect some variation across sites around x due simply to chance.

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- **Collapsed repetitive sequences**: several similar genomic regions are represented by one location in the reference, so reads from all copies map there.
- **Duplicated genes or paralogues**: reads from related gene copies incorrectly map to the same reference position.
- **Copy-number variation**: some individuals genuinely carry additional copies of the region.

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A Phred-scaled confidence that the site has been correctly called

- Score of 30 indicates a 1 in 1,000 change of an error

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Integrates information about

- **Bases observed**: how many reads support the reference and alternative alleles.
- **Base quality**: whether individual bases are likely to have been sequenced correctly.
- **Mapping quality**: whether the reads are likely to have been aligned correctly
- **Read depth**: more reads supporting an allele generally provide stronger evidence.
- **Alignment uncertainty**: BCFtools mpileup uses alignment info to reduce the influence of bases that may be misaligned, particularly around indels.
- **Genotype likelihoods**: for each individual, it calculates the likelihood of genotypes such as 0/0, 0/1 and 1/1.
- **Evidence across individuals**: BCFtools call combines the genotype likelihoods across samples and applies its population-genetic model to assess whether an alternative allele is present.

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Naively, we expect the forward and reverse strands to be sequenced the same number of times, but bias can be produced:

- **PCR bias:** some fragments may amplify more successfully in one orientation.
- **Alignment artefacts:** reads from one strand may map differently
- **Read-position effects:** the alternative allele may occur near the ends of reads from one orientation, where error rates are higher.
- **Adapter contamination or incomplete trimming:** residual adapter sequence can affect one read orientation more strongly.
- **Library-preparation biases:** fragmentation, amplification or target-capture steps can produce unequal strand representation.

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GQ < 30 | DP < 5

Sample	Original genotype
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C	1/1
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After this step it is important to remove sites with > some % missing data

How much do we lose during filtering?

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Full unfiltered dataset	1,658,076,957	1,521,844,025	99,448,272	17,240,263

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After droping sites above threshold DP -e 'INFO/DP>3750'	1,450,599,180	1,408,460,066	42,139,114	-

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After soft filtering: **Set low DP genotypes to missing** -S . -e 'FMT/DP<5 | FMT/GQ<20' **drop sites with > 0.2 missing data**	454,361,480	429,399,898	24,961,582	-

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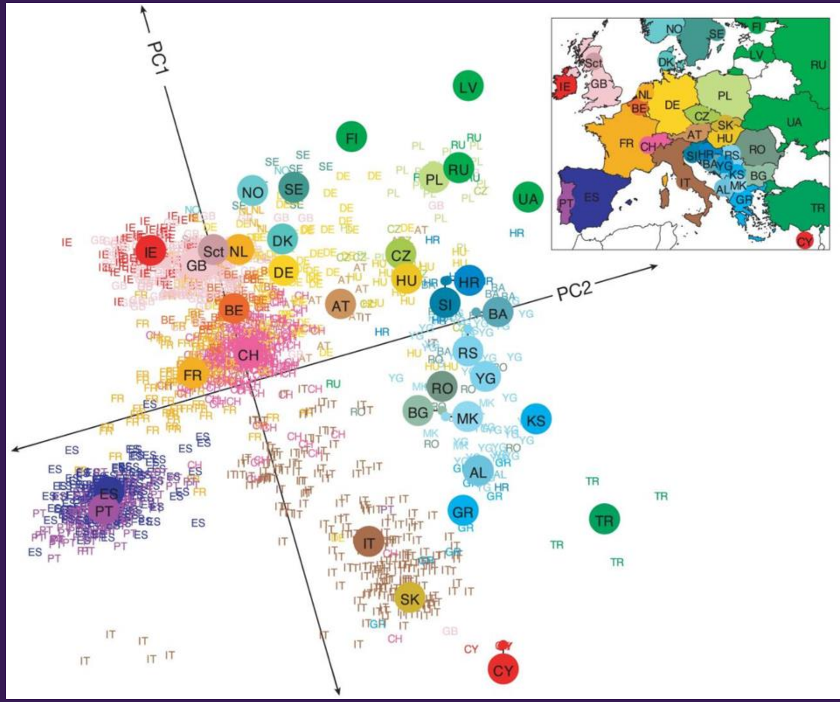
<tr><td>After additional hard and soft filtering
Drop sites below depth threshold
-e 'INFO/DP<1500'
Drop genotypes with < 10 reads
-e 'FMT/DP<10',
Drop sites with > 10% missing data
'F_MISSING > 0.1'</td><td>328,831,939</td><td>310,294,741</td><td>18,537,198</td><td>-</td></tr>
</table>

2.1 Principal components analysis



PCA can be used to summarise genomic datasets

The first analysis that most people do!

- Each SNP represents a dimension of genetic variation.
- With millions of SNPs, individuals occupy a high-dimensional "genetic space".
- PCA reduces this variation to a few axes called principal components (PCs).

How does it do?

- PCA identifies combinations of SNPs whose allele counts covary across individuals.
- For example, if individuals carrying allele A at one SNP also tend to carry particular alleles at many other SNPs, those SNPs may contribute in the same direction to a principal component.
- PCA combines these correlated allele-frequency patterns into axes.

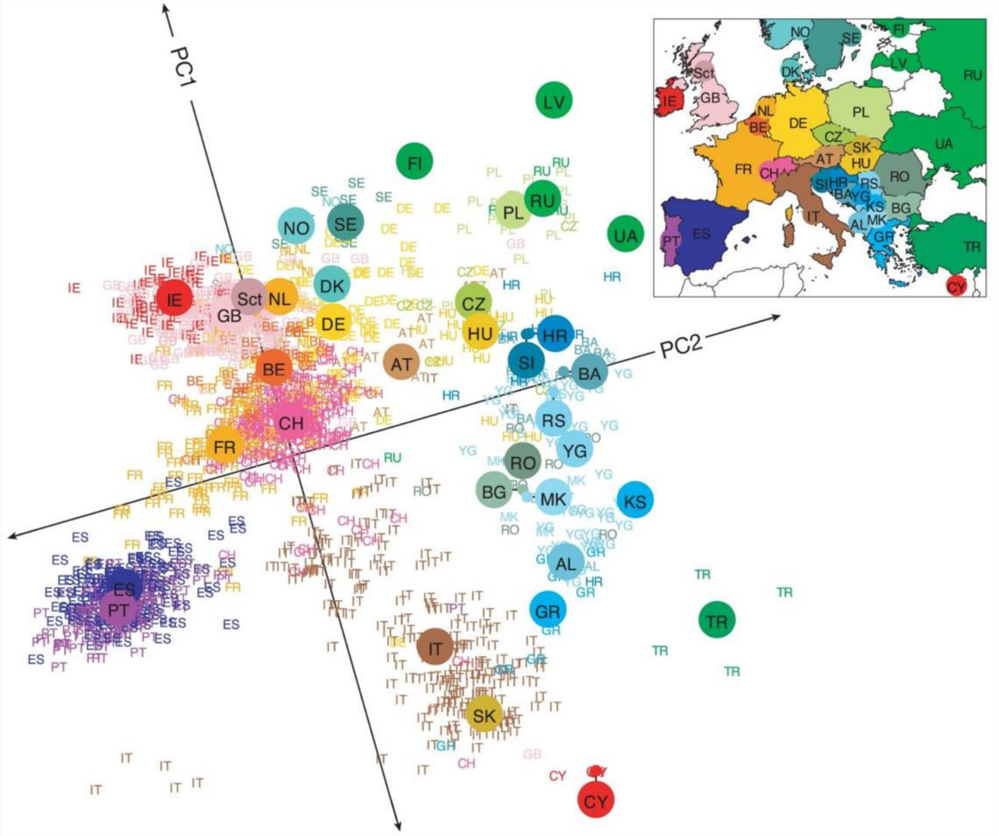
PCA can be used to summarise genomic datasets

PCA can reveal

- Major patterns of population structure
- Genetic clusters and geographic gradients
- Admixed or intermediate individuals
- Closely related, duplicated or unusual samples
- Sample mislabelling and technical batch effects

Some examples

Genes mirror geography within Europe



https://www.nature.com/articles/nature07331

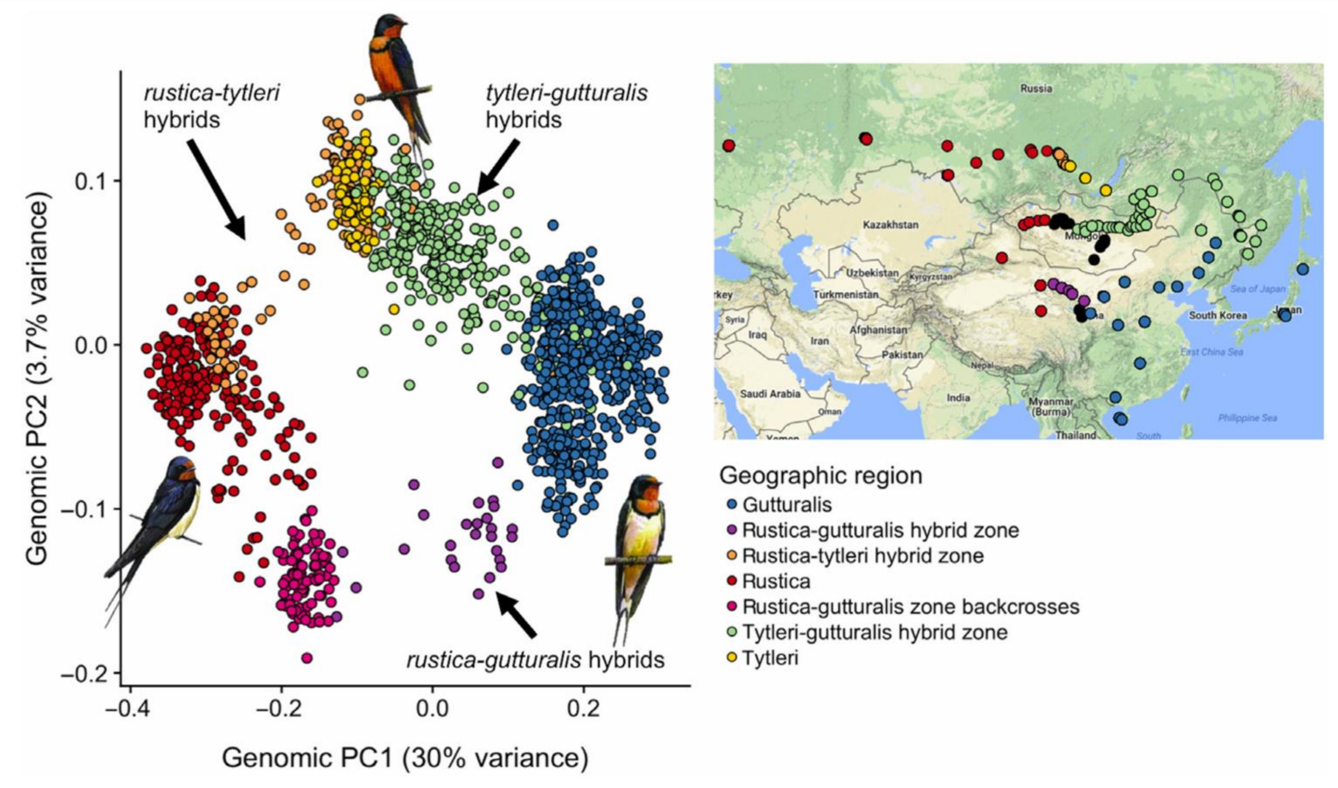
3,000 European individuals genotyped at over half a million variable DNA sites in the human genome.

Despite low average levels of genetic differentiation among Europeans, we find a close correspondence between genetic and geographic distances

indeed, a geographical map of Europe arises naturally as an efficient two-dimensional summary of genetic variation in Europeans.

Some examples

A ring around the Tibetan Plateau

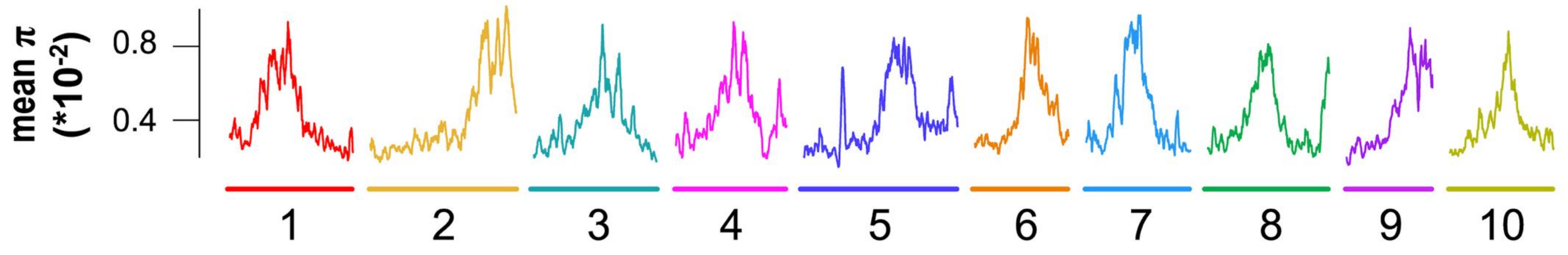


We identified three genetic clusters that corresponded to the three subspecies and varied dramatically in extent of admixture.

We found limited hybridisation between *rustica tytleri* and *rustic gutturalis,* but extensive admixture between *tytleri* and *gutturalis*

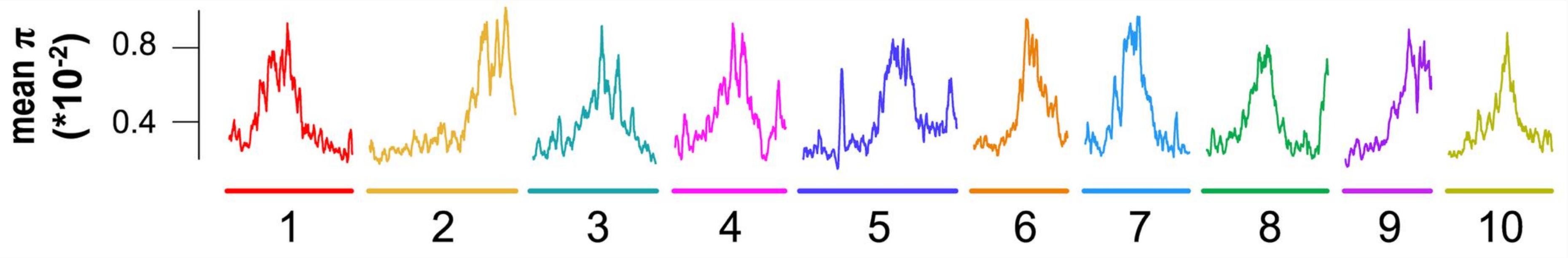
https://onlinelibrary.wiley.com/doi/full/10.1111/ele.13420

2.2 Genome scans



Genome scans

- A genome scan examines how a genetic statistic varies along the genome.
- Statistics are calculated repeatedly across chromosomes or contigs.
- Peaks and troughs identify regions that differ from the genomic background.
- Genome scans can reveal candidate regions associated with:
  - Selection
  - Reduced gene flow
  - Recombination-rate variation
  - Structural variants
  - Unusually high or low genetic diversity



Why calculate statistics in windows?

- Individual SNPs contain limited and noisy information.
- Genome scans group neighbouring sites into **windows**.
- Larger windows contain more information but give lower spatial resolution.
- Smaller windows give higher resolution but produce noisier estimates.
- Windows can contain a fixed number of base pairs or a fixed number of SNPs.

Window size: the amount of sequence included in each estimate

Step size: the distance the window moves before the next estimate

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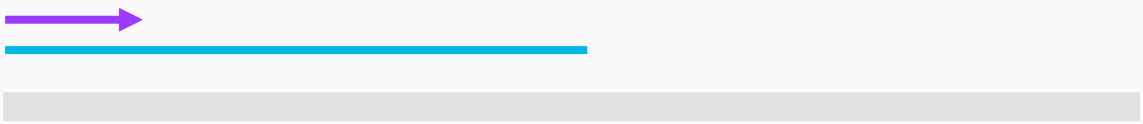

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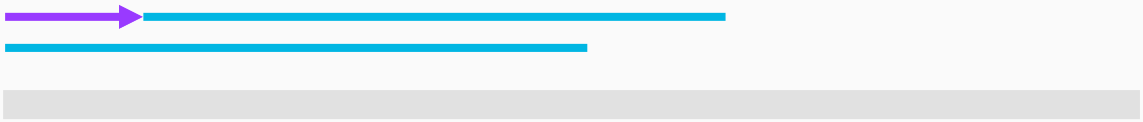
- Larger windows contain more information but give lower spatial resolution.
- Smaller windows give higher resolution but produce noisier estimates.
- Windows can contain a fixed number of base pairs or a fixed number of SNPs.



Window size: the amount of sequence included in each estimate

Step size: the distance the window moves before the next estimate

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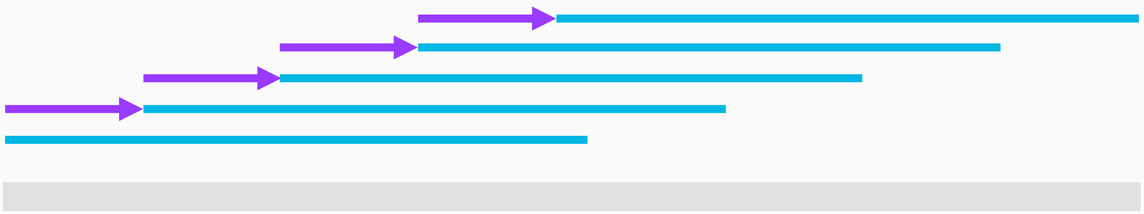
If s < w windows overlap



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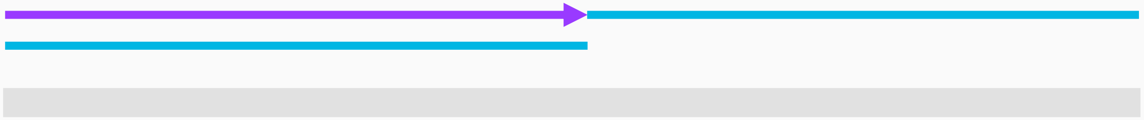
If **s** = **w** windows are non-overlapping



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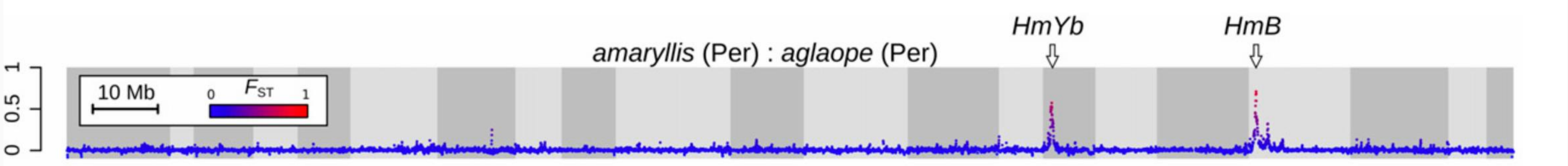
What statistics to calculate?

- Anything you want!
- We will calculate classic measures of within- and between-population variation

Statistic	What does it measure?	Interpretation
π	Mean pairwise difference between sequences within a population	Genetic diversity
d_{XY}	Mean pairwise difference between sequences from populations X and Y	Absolute divergence
F_{ST}	Differentiation relative to the diversity within populations	Relative divergence

An example

Genome-wide evidence for speciation with gene flow in *Heliconius* butterflies



We characterized patterns of divergence across the genome between populations at various levels of divergence and geographic separation using the fixation index, F_{ST}. At the earliest stage of divergence, between parapatric races, F_{ST} was low throughout the genome with just a few narrow peaks, which are partly explained by known wing pattern divergence. Between *H. m. aglaope* and *H. m. amaryllis* from Peru

https://genome.cshlp.org/content/23/11/1817